SAMamba: State Attention Mamba for Hyperspectral Image Classification

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***Abstract*—State-space models (SSMs), such as Mamba, have exhibited superior performance in long-range dependency modeling with linear complexity. However, their causal modeling paradigm, topology-disruptive serialization, and channel-independent state evolution limit their ability to model noncausal spectral–spatial dependencies in hyperspectral images (HSIs). To address these limitations, this article proposes state attention Mamba (SAMamba), a dual-branch heterogeneous state-space framework for HSI classification. Unlike existing Mamba-based methods that mainly rely on predefined scanning strategies or feature-level fusion, SAMamba explicitly intervenes in two key stages of state-space modeling: sequence construction and state evolution. In the spatial branch, the amplitude–phase cooperative semantic sequence construction (APC) module constructs semantic sequences to enhance long-range interactions among semantically correlated pixels, while the frequency-aware state-space augmentation (FAS) module compensates for the spatial structural bias introduced by semantic reordering by injecting multiscale structural priors into hidden states. In the spectral branch, the bidirectional spectral sequence construction (BSS) module constructs forward and backward spectral sequences for bidirectional spectral context modeling, while the bidirectional fusion spatial-aware state (BFS) module fuses bidirectional spectral context and spatial information in the state space to strengthen spectral–spatial joint representation. Experimental results on four public HSI benchmarks verify the effectiveness of SAMamba. Specifically, on the Indian Pines dataset, SAMamba improves over the second-best method by 0.46%, 1.56%, and 0.52% in overall accuracy (OA), average accuracy (AA), and Kappa coefficient, respectively.**

Received 12 February 2026; revised 1 April 2026 and 10 May 2026; accepted 8 June 2026. Date of publication 15 June 2026; date of current version 29 June 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62535010 and Grant 62271200, and in part by the NUIST Students' Platform for Innovation and Entrepreneurship Training Program (National-Level College Student Innovation and Entrepreneurship Training Program) under Grant 202510300078. *(Corresponding author: Bing Tu.)*

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Digital Object Identifier 10.1109/TGRS.2026.3703468

***Index Terms*—Hyperspectral image (HSI) classification, Mamba.**

I. Introduction

With the continuous advancement of imaging technologies, hyperspectral imaging has overcome the information expression limitations of traditional images through its unique acquisition capability [1]. Unlike conventional trichromatic images that contain only three discrete bands, hyperspectral images (HSIs) represent each pixel with hundreds of continuous and finely resolved spectral responses, thereby forming a 3-D spatial–spectral data cube with rich spectral signatures and spatial context [2]. Due to these characteristics, HSI has been widely applied in various domains, such as military surveillance [3], agricultural disaster response [4], [5], and mineral exploration [6]. As a fundamental task in HSI processing, HSI classification aims to accurately distinguish pixelwise categories and has attracted sustained attention in both research and application.

Traditional HSI classification methods are mainly built upon machine learning models, such as support vector machines (SVMs) [7], maximum likelihood classification, and decision trees [8]. Although these methods can achieve reasonable performance in low-dimensional settings, they rely heavily on handcrafted features and are often inadequate for handling the high dimensionality, strong redundancy, and complex nonlinear spatial–spectral correlations inherent in hyperspectral data.

Recent advances in deep learning have significantly promoted the development of HSI classification. In particular, convolutional neural networks (CNNs) [9], [10], [11] have become one of the dominant paradigms due to their strong capability in extracting local spectral–spatial patterns. However, the locality of convolution inherently limits the receptive field, making it difficult to effectively model long-range dependencies. As a result, deeper architectures are often required to capture broader contextual information, which may in turn increase the risk of overfitting under the common condition of limited labeled samples in HSI datasets.

To alleviate the restriction of local receptive fields, Transformer-based models [19], [22] have recently been introduced into HSI classification. By means of self-attention, Transformers are able to capture global contextual relationships and long-range dependencies across both spatial and spectral dimensions. Nevertheless, the quadratic computational

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and memory complexity of self-attention with respect to sequence length poses a substantial challenge for hyperspectral imagery, particularly in scenarios involving high spatial resolution and numerous spectral bands, thereby limiting the scalability and practical applicability of Transformer-based methods.

Mamba [27], as a recently proposed selective state-space model (SSM), achieves efficient modeling of long-range dependencies with linear computational complexity by introducing a selective scanning mechanism combined with hardware-aware parallel algorithms. Benefiting from its favorable efficiency–capacity tradeoff, Mamba has demonstrated outstanding performance in sequence modeling tasks, particularly in natural language processing (NLP). However, unlike linguistic data that inherently follows a 1-D sequential structure, visual data exhibit a natural 2-D spatial organization, which poses fundamental challenges for directly transferring Mamba to visual tasks. Moreover, the causal modeling paradigm adopted by Mamba further limits its ability to capture global contextual interactions in complex visual scenes. To adapt SSMs for visual perception, existing approaches—including VMamba [28], Vision Mamba [29], LocalMamba [30], and PlainMamba [31]—typically flatten 2-D feature maps into multiple 1-D sequences via multiscan strategies. Although such designs expand the receptive field and partially alleviate directional sensitivity, they inevitably disrupt the original spatial topological relationships among pixels while introducing additional computational overhead.

These limitations are particularly pronounced in HSI classification. HSI exhibits a 3-D spatial–spectral cube structure, where accurate recognition relies on the strong coupling between spatial context and continuous spectral signatures. However, existing Mamba-like HSI solutions [32] still inherit several intrinsic limitations.

1) *Topology-Breaking Serialization:* Most methods serialize 2-D spatial features to fit causal SSMs, which breaks spatial topology and introduces directional bias.
2) *Long-Range Dependency Decay:* Causal propagation causes long-range spatial dependencies to attenuate, limiting global aggregation of distant but spectrally similar regions.
3) *Incomplete Spectral Context Modeling:* The causal paradigm restricts full-context interaction across spectral bands, making continuous spectral dependencies difficult to capture.
4) *Weak Spatial–Spectral Collaboration:* Decoupled state updates limit joint representation learning between spatial structures and spectral evolution.
5) *Insufficient Spatial–Spectral Coupling:* Existing modeling pipelines cannot fully exploit the inherent spatial–spectral coupling in HSI, leading to suboptimal discrimination in complex scenes.

To address these challenges, we propose a spatial–spectral modeling framework at the state-space level, which explicitly intervenes at two critical stages: sequence construction and state evolution. Specifically, we introduce a novel state-space modeling framework termed state attention Mamba (SAMamba). SAMamba adopts a dual-branch parallel architecture to jointly model spatial and spectral information within the state space, thereby enhancing spatial and spectral representation learning and enabling more comprehensive joint representation learning.

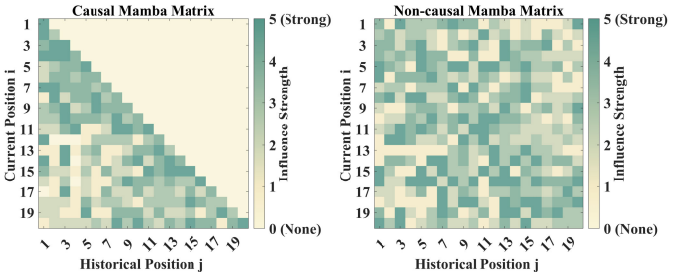


Fig. 1. (Left) Correlation matrix of causal Mamba, where causality restricts interactions to historical positions, resulting in weak long-range dependencies and disrupted spatial structures. (Right) Our SAMamba adopts a noncausal state-space design to promote global interactions, preserve spatial topology, and enhance long-range dependencies.

In terms of spatial modeling, we introduce the amplitude–phase cooperative semantic sequence construction (APC) module to enhance interactions between semantically similar pixels and mitigate the attenuation of long-range dependencies caused by causal state propagation, while preserving key local geometric cues. In addition, we incorporate the frequency-aware state-space augmentation (FAS) module into the state space to hierarchically compensate for structural distortions caused by semantic reordering (sequentialization), thereby improving spatial consistency and structural fidelity.

In spectral modeling, we introduce the bidirectional spectral sequence construction (BSS) module to construct bidirectional spectral sequences, enabling noncausal spectral context modeling and enhancing the representation of long-range spectral dependencies. Furthermore, we introduce a bidirectional fusion spatial-aware state (BFS) module in the state space to overcome the limitations of Mamba's channel-independent state update mechanism in spectral–spatial collaborative modeling. This enables explicit interaction between spectral context information and spatial structural cues, thereby further strengthening the learning of joint spatial-spectral representations.

To clearly elucidate the inherent limitations of causal SSMs in HSI classification and provide a theoretical basis for our design, we conduct a dedicated motivation analysis in Section II. This analysis systematically examines the long-range decay characteristics of causal state evolution, further emphasizing the necessity of intervening at the state-space level to achieve efficient spectral–spatial modeling.

The main contributions of this work are summarized as follows.

1) We propose the SAMamba, addressing HSI classification. By introducing a dual-branch parallel mechanism and heterogeneous state-space modeling, it effectively addresses the limitations of causal dependency modeling and supports efficient extraction of spectral–spatial coupled features in HSIs.

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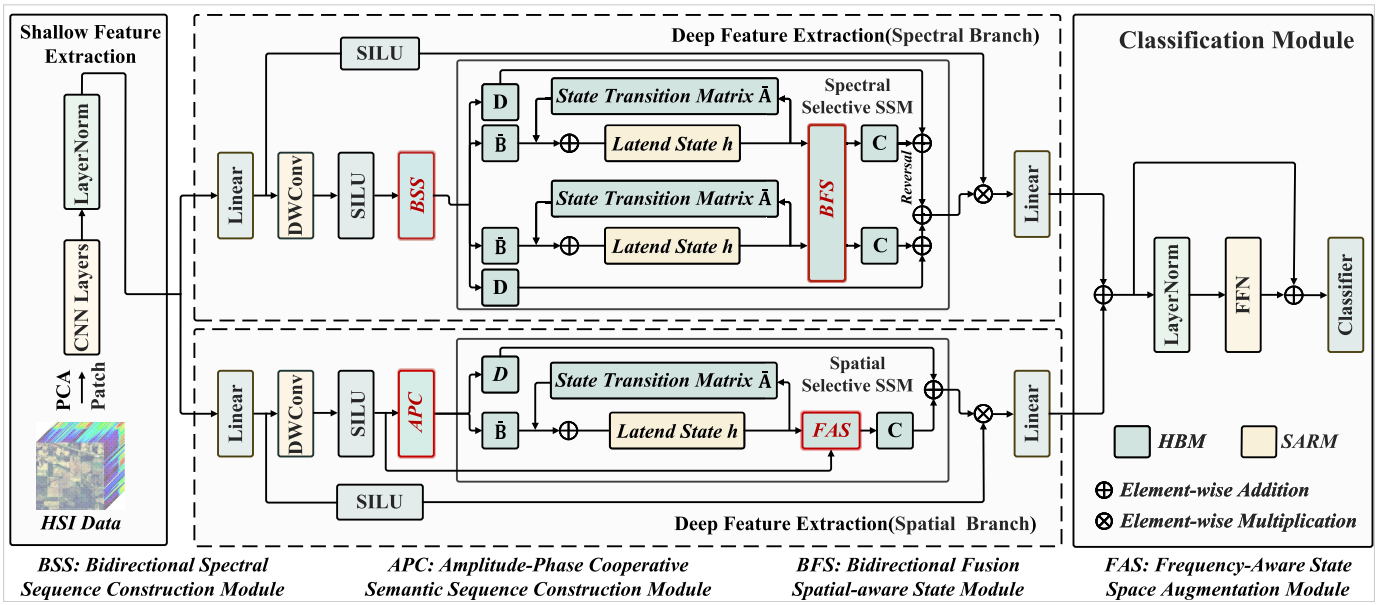


Fig. 2. This framework diagram illustrates the core of SAMamba. CNN Layers extract shallow features, while the spectral–spatial dual-branch structure and heterogeneous turntable transfer mechanism capture deep features. The spatial branch uses the APC module to model semantic sequences and the FAS module to enhance spatial structure. The spectral branch applies the BSS module to construct bidirectional spectral sequences and the BFS module to fuse bidirectional states and spatial information. The fused features are further processed by an FFN and classified to produce the final result.

2) In the spatial branch, we design the APC module to construct semantic-guided sequences that enhance global spatial interaction and the FAS module to supplement spatial structural information, collectively improving both semantic relevance and structural fidelity in spatial modeling.
3) In the spectral branch, we introduce the BSS module to construct bidirectional spectral sequences for causality compensation and the BFS module to enable bidirectional state fusion with spatial perception, thereby enhancing joint spectral–spatial evolution.
4) Rigorous benchmarking on public datasets demonstrates that the proposed dual-branch coordination and heterogeneous state transition modeling strategy exhibits significant performance advantages over existing state-of-the-art methods.

The structure of this article is outlined as follows. Following a review of existing studies in Section III, the introduced approach is elaborated in Section IV. Section V provides the experimental configuration, outcomes, and corresponding discussions. Section VI outlines the key conclusions of this work and discusses potential extensions for future research.

II. Motivation

A. Long-Range Dependency Attenuation in Causal State-Space Modeling

Based on the causal state-space formulation of Mamba, the state update and output can be written as

$$h_t = \bar{\mathbf{A}} h_{t-1} + \bar{\mathbf{B}} x_t, \quad y_t = \mathbf{C} h_t + \mathbf{D} x_t \tag{1}$$

where the output at time step k is given by

$$y_k = \sum_{i=0}^{k} \mathbf{C} \bar{\mathbf{A}}^{k-i} \bar{\mathbf{B}} x_i + \mathbf{D} x_k. \tag{2}$$

Under this formulation, the contribution of a past input x_i to the current output y_k is determined by $\bar{\mathbf{A}}^{k-i}$. To ensure numerical stability, the transition matrix $\bar{\mathbf{A}}$ is typically constrained to have a spectral radius smaller than one. As the distance $(k-i)$ increases, $\bar{\mathbf{A}}^{k-i}$ gradually decays, causing long-range dependencies to weaken during recursion.

This attenuation is an intrinsic property of causal SSMs under stability constraints. Although suitable for strictly causal sequence modeling, it becomes restrictive for HSI classification. HSI data often contain strong long-range spectral correlations and nonlocal semantic dependencies, where distant regions can still be highly related. However, under causal state evolution, such information is progressively weakened and becomes difficult to preserve in the hidden state. Fig. 1 further shows that causal Mamba not only limits long-range interactions but also disrupts the original spatial structure.

Furthermore, the attenuation of long-range dependencies is closely related to the order of input. When correlated elements are placed far apart after being serialized, their interactions during causal state propagation are further weakened. Therefore, for HSI data, a simple 1-D unfolding sequence is insufficient, as it may disrupt spatial topological structures, spectral correlations, and nonlocal semantic dependencies. This highlights the importance of sequence construction, as it determines the effective interaction distance between relevant spatial or spectral elements prior to state evolution.

B. Why State-Space Intervention Is Necessary for Effective Global Modeling?

SSMs recursively compress historical information into hidden states, which serve as the core carrier of global context. Therefore, the quality of global representations largely depends on how the hidden state is constructed and evolved.

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Prior studies have shown an intrinsic connection between linear attention and causal SSMs. Self-attention aggregates context through explicit pairwise interactions, whereas SSMs compress historical information into hidden states through recursive updates and retrieve it through an observation mapping. From this perspective, the observation mapping can only read information already stored in the hidden state, rather than introduce new context. As a result, global modeling ability is fundamentally governed by the state evolution process itself.

This issue becomes more pronounced in HSI modeling. Although sequence construction can reduce the effective distance between correlated elements, the recursive state evolution process still determines how such information is accumulated and preserved in hidden states. In particular, serialization may introduce structural bias, while causal propagation and channel-independent state updates can further aggravate information fragmentation and weaken spectral–spatial interaction. Since these limitations are accumulated inside the state space during recursion, they are difficult to fully compensate for through later feature-level operations.

Therefore, beyond the sequence construction stage discussed above, effective HSI global modeling also requires explicit intervention in state evolution. State-space operations before observation mapping can directly modulate hidden states, helping to preserve structural information, reduce fragmented context propagation, and enhance spectral–spatial collaborative representation. Thus, sequence construction and state evolution constitute two critical stages for improving causal state-space modeling in HSI classification.

III. Related Works

A. HSI Classification Based on CNN

In the era of deep learning, CNNs have emerged as a predominant approach for HSI classification [12], [13]. Early CNN-based methods primarily focus on jointly modeling spectral and spatial information through hierarchical convolutional operations.

Zhong et al. [14] proposed the spectral–spatial residual network (SSRN), which introduces residual learning for end-to-end spectral–spatial feature extraction. Paoletti et al. [15] developed a deep and dense CNN architecture to enhance feature reuse through dense interlayer connections, and further proposed deep pyramidal residual networks [16] to enrich high-level representations. Xu et al. [17] introduced the spectral–spatial unified network (SSUN), which combines band-grouped LSTMs with multiscale 2D-CNNs for unified spectral–spatial modeling.

Overall, CNN-based methods are effective in extracting local spectral–spatial representations from HSI data due to their strong inductive biases. However, their locality-driven design limits their ability to capture long-range contextual dependencies in complex HSI scenes.

B. HSI Classification Based on Transformer

Transformer architectures, leveraging self-attention mechanisms, have recently emerged as a powerful alternative for HSI classification by enabling explicit modeling of long-range spectral–spatial dependencies [20], [21]. Compared with CNNs, Transformers offer superior global contextual representation capability.

Hong et al. [23] proposed SpectralFormer, which employs grouped spectral embedding and cross-layer adaptive fusion for spectral context modeling. Peng et al. [24] developed the cross-attention spatial–spectral Transformer (CASST) to jointly capture spatial and spectral dependencies. Sun et al. [25] proposed the spectral–spatial feature tokenization Transformer (SSFTT), which combines CNN-based feature extraction with tokenization for discriminative semantic representation. Li et al. [26] introduced SWFormer, which uses stochastic window-based attention and dual-plane interaction to balance local modeling and computational efficiency.

Despite their strong global modeling capability, Transformer-based HSI classification methods still suffer from high computational cost on high-dimensional HSI data. In addition, patchwise tokenization or window-based attention may weaken fine-grained spatial–spectral continuity. These limitations motivate more efficient sequence modeling frameworks for HSI classification.

C. HSI Classification Based on Mamba

Leveraging its capability for long-range dependency modeling with linear computational complexity, Mamba has recently attracted increasing attention in HSI classification, particularly for handling high-dimensional spectral sequences [33], [34]. Existing Mamba-based HSI classification methods can be broadly categorized into three types according to their primary design focus: spectral-dominant modeling, spatial–spectral scanning and fusion, and hybrid architecture-based methods.

Spectral-dominant methods mainly exploit Mamba's sequential modeling capability along the spectral dimension. For example, SpectralMamba [35] models spectral dependencies through segmented sequence scanning and gated fusion. Although such methods can effectively capture continuous spectral correlations, their limited use of spatial context weakens the modeling of intrinsic spectral–spatial coupling.

Spatial–spectral scanning and fusion methods further design spatial scanning, spectral scanning, and fusion mechanisms for HSI modeling. MambaHSI [36] uses spatial and spectral Mamba blocks to model sequentially flattened pixel-level spatial sequences and grouped spectral sequences, respectively, followed by adaptive spatial–spectral fusion. SpiralMamba [37] introduces forward and reverse spiral scans to preserve spatial continuity and relative positional relationships around the central pixel. IGroupSS-Mamba [38] employs an interval group S6 mechanism with multidirectional scanning and cascaded spatial–spectral operators for efficient spatial–spectral modeling. S^2Mamba [41] designs patch cross scanning and bidirectional spectral scanning, together with a spatial–spectral mixture gate for adaptive fusion.

Hybrid architecture-based Mamba methods enhance feature representation by introducing complementary structures such as Transformers, CNNs, or attention mechanisms. For example, HSI-MFormer [39] uses Transformer modules

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for multiscale context modeling, while DualMamba [40] adopts a parallel Mamba–CNN architecture to combine long-range dependency modeling and local feature extraction. However, these methods mainly rely on external architectural complementarity, while the sequence construction and state evolution mechanisms of Mamba remain insufficiently improved.

Although the aforementioned methods enhance Mamba's ability to model the spatio-spectral information in HSIs from various perspectives, these improvements primarily focus on scan design, external structure enhancement, and feature fusion following state evolution. With regard to spatial modeling, serializing 2-D spatial structures into 1-D representations introduces artificial sequence priors, which may disrupt the original spatial topological relationships and weaken interactions between semantically related but spatially disconnected pixels. Furthermore, spatial–spectral information or data obtained from multidirectional scans is typically integrated only after independent state evolution, resulting in a lack of direct control over the hidden state update process. Consequently, the structural bias introduced by spatial serialization, along with the limitations of Mamba's causal state propagation and channel-agnostic state update mechanisms, remain inadequately addressed in existing methods. For example, although SpiralMamba improves spatial continuity through spiral scanning, it still relies on predefined scan paths and does not explicitly bring semantically similar pixels closer during sequence construction. S^{2}Mamba strengthens bidirectional spectral context modeling and spectral–spatial fusion, but such interactions are mainly performed after state evolution rather than within the state space itself. In contrast, SAMamba explicitly intervenes in the Mamba model at two critical stages—sequence construction and state evolution—to achieve more robust long-range interactions, structural compensation, and spatio-spectral collaborative modeling.

IV. Proposed Network

A. Preliminaries

SSMs serve as an important tool for describing classical linear time-invariant (LTI) systems. Their core mechanism realizes the mapping from input sequences to output sequences through internal latent states. For continuous-time LTI-SSMs, the process is jointly described by the state transition and observation equations

$$\begin{aligned} h'(t) &= \mathbf{A}\cdot h(t) + \mathbf{B}\cdot x(t) \\ y(t) &= \mathbf{C}\cdot h(t) \end{aligned} \tag{3}$$

where $h(t) \in \mathbb{R}^N$ denotes the system state vector at time t, $x(t)$ is the input signal, and $y(t)$ is the output signal. Here, $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B} \in \mathbb{R}^{N\times 1}$, and $\mathbf{C} \in \mathbb{R}^{1\times N}$ denote the state transition, input, and output matrices, respectively.

To bridge continuous SSMs with deep learning architectures that operate on discrete sequences, temporal discretization is required. By employing the zero-order hold (ZOH) approach and introducing a discretization step Δ, the continuous parameters $(\mathbf{A}, \mathbf{B})$ can be transformed into their discrete counterparts $(\bar{\mathbf{A}}, \bar{\mathbf{B}})$

$$\begin{aligned} \bar{\mathbf{A}} &= \exp(\Delta\cdot\mathbf{A}) \\ \bar{\mathbf{B}} &= (\Delta\cdot\mathbf{A})^{-1}\cdot\left(\exp(\Delta\cdot\mathbf{A}) - \mathbf{I}\right)\cdot\Delta\cdot\mathbf{B} \end{aligned} \tag{4}$$

where $\mathbf{I}$ denotes the identity matrix. The discretized SSM is computed in a linear recursive form

$$\begin{aligned} h_t &= \bar{\mathbf{A}}\cdot h_{t-1} + \bar{\mathbf{B}}\cdot x_t \\ y_t &= \mathbf{C}\cdot h_t + \mathbf{D}\cdot x_t. \end{aligned} \tag{5}$$

Models constructed based on such discretized, time-invariant SSMs are referred to as S4 models. During training, the above linear recursive process can be equivalently represented as a global convolution between the input sequence $x \in \mathbb{R}^L$ and a structured convolution kernel $\bar{\mathbf{K}} \in \mathbb{R}^L$

$$y = x * \bar{\mathbf{K}} \tag{6}$$

where $\bar{\mathbf{K}} = [\mathbf{CB}, \mathbf{CAB}, \mathbf{CA}^2\mathbf{B}, \ldots, \mathbf{CA}^{L-1}\mathbf{B}]$ and L is the sequence length.

In S4 models, the core parameters $\mathbf{A}, \mathbf{B}$, and $\mathbf{C}$, and the discretization step Δ are fixed and independent of the input sequence x. This time-invariant property limits the model's ability to adapt its dynamics to different inputs.

The Mamba model overcomes this limitation by introducing a *selection mechanism*, which dynamically parameterizes the key components of the SSM as functions of the input x. Specifically, Mamba realizes parameter dynamization through

$$\begin{aligned} \mathbf{\Delta} &= s_\Delta(x) \\ \mathbf{B} &= s_\mathbf{B}(x) \\ \mathbf{C} &= s_\mathbf{C}(x) \end{aligned} \tag{7}$$

where s_Δ, $s_\mathbf{B}$, and $s_\mathbf{C}$ are simple linear projection layers.

These input-dependent parameters are then used to compute the discretized parameters $(\bar{\mathbf{A}}, \bar{\mathbf{B}})$ and perform recursive state updates. As a result, Mamba constructs an input-dependent, time-varying SSM with content-aware modeling capability.

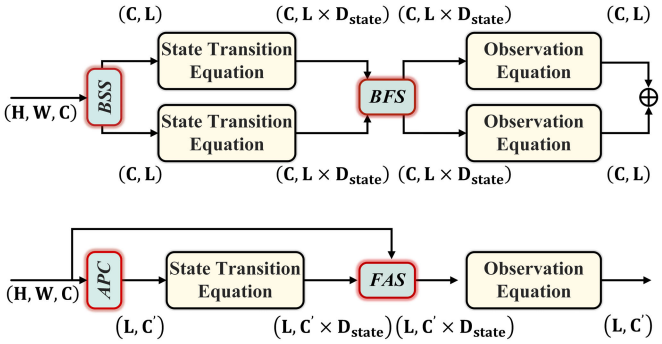


Fig. 3. Data flow diagram of APC, FAS, BSS, and BFS modules in the spatial and spectral branches.

B. Overall Architecture

The overall architecture of the proposed framework is shown in Fig. 2, with the detailed data flow illustrated in Fig. 3. The framework adopts a dual-branch architecture comprising

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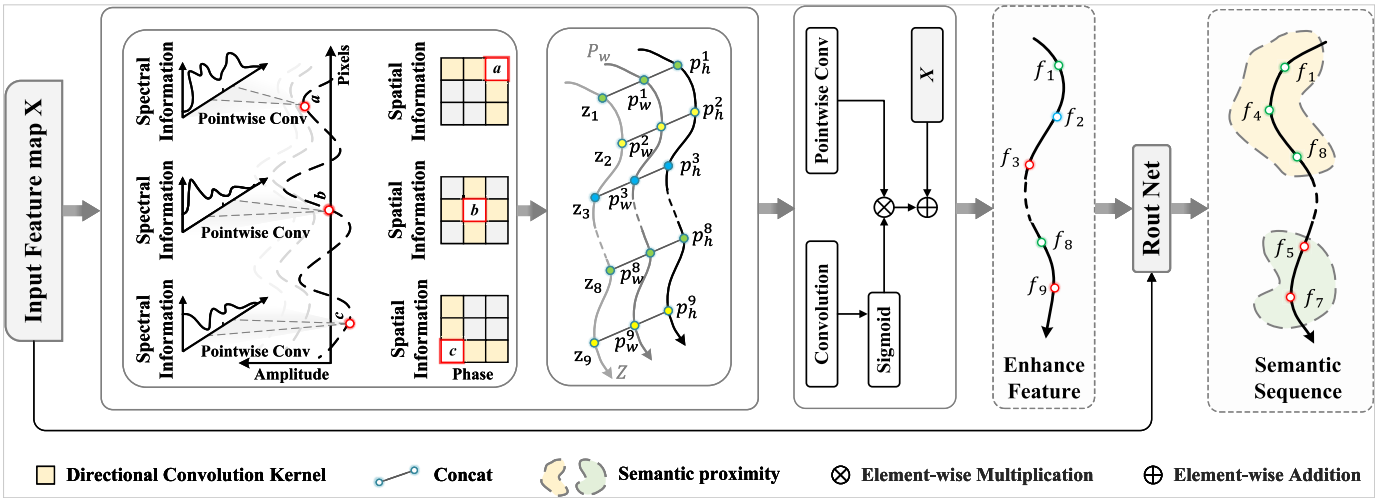


Fig. 4. Detailed mechanism diagram of the APC module.

a spatial branch and a spectral branch, both built upon a shared Mamba state-space modeling backbone. Each branch incorporates specialized modules for sequence construction and state-space processing to capture complementary spatial and spectral information.

1) Spatial Branch: The spatial branch includes semantic sequence construction, state-space modeling, and state augmentation. The APC module first reorganizes the input feature map into a semantic sequence, which is processed by the Mamba state transition equation to obtain spatial hidden states. The FAS module then fuses these states with spatially preserved shallow features in the state space, and the augmented states are projected by the Mamba observation equation to produce the spatial output.

2) Spectral Branch: The spectral branch focuses on bidirectional spectral modeling and state-space fusion. The BSS module constructs forward and backward spectral sequences, which are independently processed by Mamba state transition equations. The BFS module fuses the bidirectional hidden states while incorporating spatial information. The fused states are projected through the observation equations, and the forward and backward outputs are summed to form the spectral output.

The four modules are selected according to two key intervention stages of state-space modeling: sequence construction and state evolution. In the spatial branch, APC addresses the dependency attenuation and semantic discontinuity caused by naive spatial serialization, while FAS compensates for the spatial structural bias introduced by semantic reordering at the hidden-state level. In the spectral branch, BSS alleviates the one-sided context limitation of causal spectral scanning, while BFS further enables bidirectional state interaction and spatial-aware spectral–spatial collaboration before readout. Therefore, these modules are not independently stacked components, but form two coupled designs: APC–FAS for semantic spatial modeling with structural compensation and BSS–BFS for noncausal spectral modeling with state-level fusion.

Finally, the outputs of both branches are combined and fed into a feed-forward network and a classifier to generate the final prediction.

C. APC Module

As discussed in Section II, causal state-space modeling suffers from long-range dependency attenuation, while naive raster serialization may place spectrally similar or spatially related pixels far apart, weakening global semantic interactions in hyperspectral imagery. To address this issue, we seek a semantics-aware sequence construction that brings spectrally similar and spatially coherent pixels closer in the sequence. Since neither raster order nor purely similarity-driven reordering can effectively balance local structure and semantic grouping, the APC module adopts an amplitude–phase coordinated design: amplitude-dominant features guide semantic aggregation, while phase-like periodic cues provide geometry-aware constraints to reduce local structural distortion. The overall design of APC is illustrated in Fig. 4.

1) Amplitude–Phase Feature Extraction and Fusion: Using a dual-branch structure, this module independently extracts spectral amplitude and spatial phase features from the input $X \in \mathbb{R}^{H\times W\times C}$, where H, W, and C denote the spatial height, width, and spectral channel number.

Amplitude branch focuses on spectral information and extracts dominant spectral features via a 1×1 convolution

$$Z = \mathrm{Conv}_{1\times 1}(X) \tag{8}$$

where $Z \in \mathbb{R}^{H\times W\times C'}$ is the spectral amplitude feature map.

Phase branch captures spatial structure and contextual information through periodic normalization and direction-aware convolution

$$\begin{aligned} \Theta &= \mathrm{Mod}(\Theta_{\mathrm{raw}}, 2\pi) \times \lambda \\ P_h &= \mathrm{Conv}_{\text{dw-h}}(\Theta), \quad P_w = \mathrm{Conv}_{\text{dw-w}}(\Theta) \end{aligned} \tag{9}$$

where λ is the scaling factor, $\Theta \in \mathbb{R}^{H\times W\times C'}$ denotes the normalized phase map, and $P_h, P_w \in \mathbb{R}^{H\times W\times C'}$ represent the horizontal and vertical phase features, respectively.

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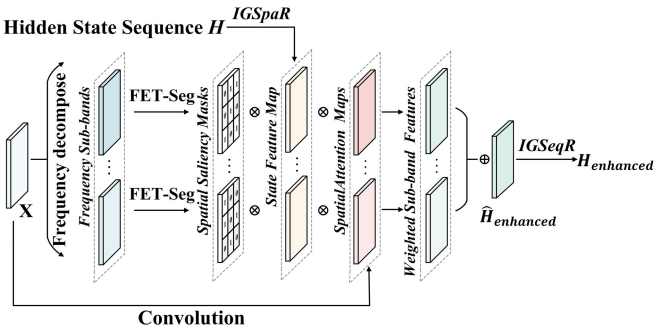


Fig. 5. Detailed mechanism diagram of the FAS module.

Feature fusion integrates amplitude and phase features through concatenation and gating

$$\begin{aligned} F_{\text{fused}} &= \text{Conv}\left([P_h, P_w, Z]\right) \\ G &= \sigma\left(\text{Conv}\left(F_{\text{fused}}\right)\right) \\ F &= F_{\text{fused}} \odot G + X \end{aligned} \tag{10}$$

where $F \in \mathbb{R}^{H\times W\times C'}$ is the enhanced feature map integrating spectral and spatial information.

Compared with directly using raw features for semantic modeling, amplitude–phase decomposition provides a more decoupled representation of spectral–spatial information, offering more discriminative support for semantic clustering and sequence construction.

2) Semantically Guided Sequence Reordering: Based on the fused feature map F, a lightweight routing network generates membership weights for each spatial position over N semantic clusters

$$R = \text{Softmax}\left(\text{Conv}_{1\times 1}\left(F\right)\right). \tag{11}$$

The cluster label for each location is obtained by

$$c_{h,w} = \arg\max\left(R\left[h, w, :\right]\right). \tag{12}$$

Subsequently, all spatial positions are sorted in ascending order by their cluster label $c_{h,w}$, generating an index list I. This operation places semantically similar pixels adjacently in the final sequence, thereby enhancing semantic locality for subsequent Mamba modeling. The reordered semantic sequence $Y = [f_1, \ldots, f_L] \in \mathbb{R}^{L\times C'}$ is then fed into the Mamba model.

D. FAS Module

Although APC enhances global semantic interactions through semantics-aware sequence rearrangement, such reordering inevitably perturbs the original spatial neighborhood relationships. Combined with 1-D serialization and causal state evolution, this disturbance can accumulate in hidden states and eventually appear as spatial structural bias in the reconstructed 2-D feature maps. Since these deviations arise during state evolution, post hoc correction after observation mapping is often insufficient.

Furthermore, frequency-domain representations describe spatial structures in a scale-separable manner: low-frequency components correspond to global layout, while high-frequency components capture local edges and texture details. Compared with the spatial domain, the frequency domain provides more direct structural cues across different scales. Based on this insight, we introduce the FAS module, which uses the frequency response of the input image as a multiscale structural anchor to guide hidden-state organization during state evolution. In this way, FAS improves structural consistency in spatial reconstruction while mitigating the structural bias introduced by APC. The architecture of the proposed FAS module is illustrated in Fig. 5.

1) State-Space Modeling and Structural Bias: The semantic sequence Y undergoes recursive scanning via the Mamba SSM. Since Mamba maintains channelwise-independent state evolution, each input channel is associated with a D_{state}-dimensional state, resulting in an effective hidden representation dimension $D = C' \cdot D_{\text{state}}$. Accordingly, the state update at step m can be written as

$$\begin{aligned} h_m^{(c)} &= h_{m-1}^{(c)} \cdot \bar{\mathbf{A}}_{\text{spa}}^{(c)} + f_m^{(c)} \cdot \bar{\mathbf{B}}_{\text{spa}}^{(c)} \\ &\left(m = 1, 2, \ldots, L;\ c = 1, 2, \ldots, C'\right) \end{aligned} \tag{13}$$

where $f_m^{(c)} \in \mathbb{R}$ denotes the input feature of the cth channel at the mth spatial position, and $h_m^{(c)} \in \mathbb{R}^{D_{\text{state}}}$ denotes the corresponding hidden state after scanning to the mth spatial position. By concatenating the hidden states of all channels at step m, we obtain $h_m = [h_m^{(1)}, h_m^{(2)}, \ldots, h_m^{(C')}] \in \mathbb{R}^D$, where $D = C' \cdot D_{\text{state}}$. Accordingly, the hidden state sequence is represented as $\mathcal{H} = [h_1, h_2, \ldots, h_L] \in \mathbb{R}^{L\times D}$.

Subsequently, via the index-guided spatial reconstruction (IGSpaR) operation, the hidden state sequence is reconstructed into a 2-D spatial structure using a cached index list

$$\hat{\mathcal{H}} = \text{Reshape}\left(\mathcal{H}\left[\text{argsort}\left(I\right)\right], (H, W, D)\right) \tag{14}$$

where $\hat{\mathcal{H}} \in \mathbb{R}^{H\times W\times D}$ denotes the reconstructed 2-D state feature map. However, prioritizing semantic information introduces structural bias, causing its frequency distribution to mismatch the original input X.

2) Frequency Subband Decomposition and Spatial Alignment: To address this bias, we perform frequency transformation and multiscale decomposition on the original input

$$\begin{aligned} X_F &= \mathcal{F}\left(X\right) \\ X_{F,k} &= X_F \odot M_k \\ X_k &= \mathcal{F}^{-1}\left(X_{F,k}\right) \end{aligned} \tag{15}$$

where the mask set $\{M_k \mid k \in K\}$ defines different frequency band regions. Each X_k represents the kth frequency subband signal in the spatial domain.

For frequency alignment, we compute an energy threshold and generate a binary mask via the frequency subband energy threshold segmentation (FET-Seg) operation

$$S_k\left(i, j\right) = \mathbb{I}\left(\left|X_k\left(i, j, :\right)\right|_1 > \gamma \cdot \mathbb{E}_{i,j}\left[\left|X_k\left(i, j, :\right)\right|_1\right]\right) \tag{16}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function and γ is a hyperparameter controlling the activation threshold. The resulting binary mask S_k is referred to as the spatial saliency mask of subband k.

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To correct the frequency distribution bias in the state feature map $\hat{H}$, we decompose it into subband components aligned with the input frequency characteristics

$$\hat{\mathcal{H}}_k = \hat{\mathcal{H}} \odot S_k. \tag{17}$$

3) Dynamic Weighted Fusion and Final Output: To inject spatial structural information from the original image into the state features, we generate spatially adaptive weights from the input and apply them to each frequency subband

$$\begin{aligned} A_k(i,j) &= \sigma\left(\mathrm{Conv}_k\left(X(i,j,:)\right)\right) \\ \hat{\mathcal{H}}_{\text{enhanced}}(i,j,:) &= \sum_{k\in K} A_k(i,j)\cdot \hat{\mathcal{H}}_k(i,j,:). \end{aligned} \tag{18}$$

This fusion enables the model to adaptively weight each frequency component according to the spatial characteristics of the input, thereby recovering structural information lost in the semantic sequence.

Then, using the index-guided sequence reconstruction (IGSqeR) operation, the enhanced spatial features are reordered back into a sequence

$$\begin{aligned} \mathcal{H}_{\text{enhanced}} &= \text{reshape}\left(\hat{\mathcal{H}}_{\text{enhanced}}[I], (L, D)\right) \\ &= \left[\hat{h}_1, \hat{h}_2, \ldots, \hat{h}_L\right]. \end{aligned} \tag{19}$$

Operations are then performed at each position of this reconstructed sequence via the Mamba observation equation to obtain the final output

$$\begin{aligned} y_m &= \hat{h}_m \cdot \mathbf{C}_{\text{spa}} + f_m \cdot \mathbf{D}_{\text{spa}} \quad (m = 1, 2, \ldots, L) \\ Y_{\text{out}} &= \left[y_1, y_2, \ldots, y_L\right] \in \mathbb{R}^{L\times C'}. \end{aligned} \tag{20}$$

Discussion on the Difference From FFT/DCT-Based Methods: Although FAS involves frequency-domain analysis, its objective and intervention stage are fundamentally different from existing FFT/DCT-based methods. Most frequency-based methods perform token mixing or attention computation directly in the frequency domain, where frequency components participate in semantic feature learning.

In contrast, FAS does not use frequency-domain transformation for representation modeling. Instead, frequency decomposition is introduced as an auxiliary structural prior to characterize multiscale spatial organization and diagnose structural bias induced by semantic reordering. More importantly, FAS is embedded into the state-space evolution process, providing spatially adaptive structural compensation on hidden states before observation mapping, rather than applying post hoc frequency attention at the feature level. This enables structure-consistent reconstruction that conventional FFT/DCT attention mechanisms cannot achieve.

E. BSS Module

In HSI classification, discriminative spectral representation depends on holistic dependencies across the full spectrum rather than only unidirectional propagation. However, causal scanning restricts the aggregation of complete spectral context. To address this issue, we construct a spectral-dominant forward sequence and a backward sequence, enabling more sufficient and symmetric spectral dependency modeling while preserving the efficiency of state-space modeling. The architecture of the proposed BSS module is illustrated in Fig. 6.

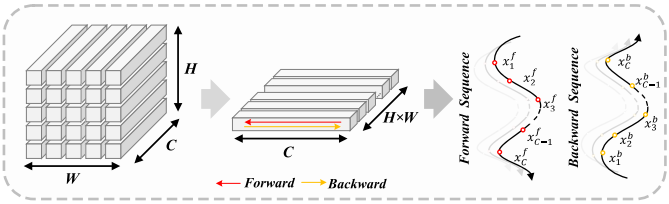


Fig. 6. Detailed mechanism diagram of the BSS module.

Specifically, the input feature map $X \in \mathbb{R}^{H\times W\times C}$ is reshaped by prioritizing the spectral dimension while flattening spatial pixels. In this way, the forward sequence is formulated as

$$X^f = \left[x_1^f, x_2^f, \ldots, x_C^f\right] \in \mathbb{R}^{C\times L} \tag{21}$$

where $x_c^f \in \mathbb{R}^{1\times L}$ denotes the spatially flattened representation of the cth spectral channel in the forward sequence, and $L = H \cdot W$.

To compensate for the limitation of unidirectional causality, the backward sequence is constructed by applying the reversal operation along the spectral dimension

$$X^b = \text{Reversal}\left(X^f\right) = \left[x_1^b, x_2^b, \ldots, x_C^b\right] \in \mathbb{R}^{C\times L}. \tag{22}$$

The forward sequence models spectral dependencies in the original spectral order, whereas the backward sequence captures complementary spectral context in the reversed order. Therefore, BSS provides bidirectional spectral inputs for subsequent hidden state evolution while preserving spectral dominance.

F. BFS Module

In hyperspectral imagery, spatial–spectral structures are jointly encoded by spectral sequences and channel dimensions, whereas channelwise-independent state evolution makes such cross-channel spatial information difficult to integrate within the state space. Moreover, although BSS introduces bidirectional spectral sequences, the forward and backward contexts are still accumulated as separate state streams, lacking effective collaboration in the global memory. To address this issue, we propose the BFS module, which performs spatially aware fusion directly on hidden states before observation mapping. In this way, BFS enables more effective bidirectional collaboration and cross-channel spatial interaction within the state space. The bidirectional fusion workflow of the proposed BFS module is illustrated in Fig. 7.

Given the forward and backward sequences, the hidden states evolve independently as

$$\begin{aligned} h_c^f &= \bar{\mathbf{A}}^f h_{c-1}^f + \bar{\mathbf{B}}^f x_c^f \\ h_c^b &= \bar{\mathbf{A}}^b h_{c-1}^b + \bar{\mathbf{B}}^b x_c^b \end{aligned} \tag{23}$$

where h_c^f denotes the hidden state at the cth spectral step in the forward hidden state sequence, and h_c^b denotes the hidden state at the cth spectral step in the backward hidden state sequence. These hidden states constitute $\mathcal{H}^f = [h_1^f, h_2^f, \ldots, h_C^f]$ and

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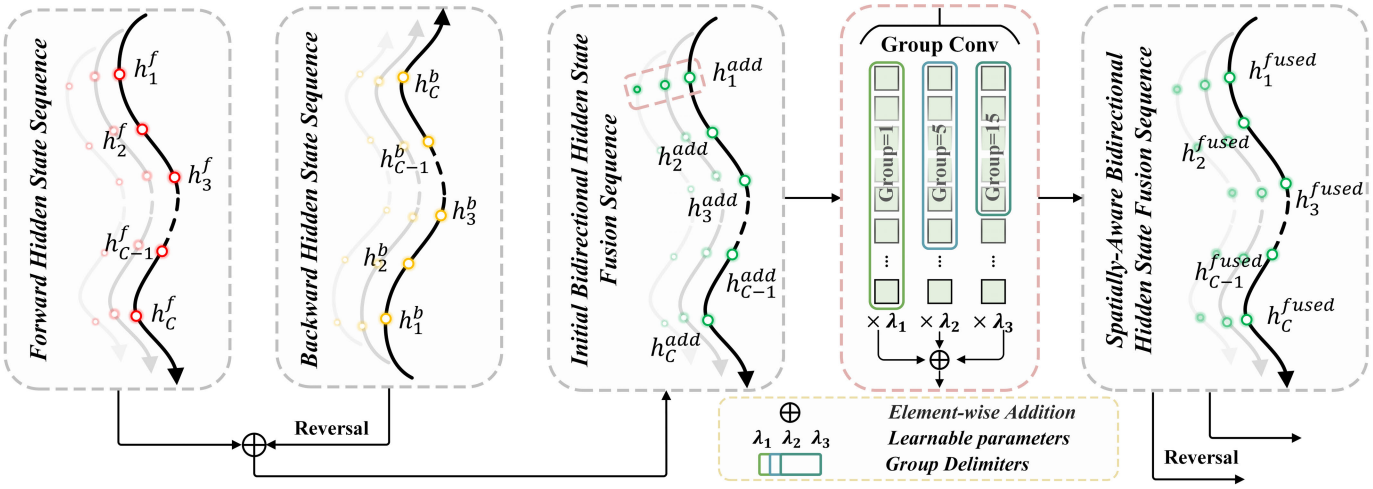


Fig. 7. Detailed mechanism diagram of the BFS module.

$\mathcal{H}^b = [h_1^b, h_2^b, \ldots, h_C^b]$, respectively. Following the channelwise state setting of Mamba, each hidden state is represented as $h_c^f, h_c^b \in \mathbb{R}^D$ with $D = L \cdot D_{\text{state}}$, where L is the number of spatial positions.

To achieve the co-evolution of spatial information and bidirectional spectral context, BFS performs bidirectional fusion at the hidden-state level. Specifically, $\mathcal{H}^b$ is first aligned with $\mathcal{H}^f$ via reversal. Then, the paired states h_c^f and h_{C-c+1}^b are fused by elementwise addition to form the initial bidirectional hidden state fusion sequence, which is then enhanced by a lightweight spatial fusion module

$$\begin{aligned} h_c^{\text{fused}} &= \text{SpatialFusion}\left(h_c^f + h_{C-c+1}^b\right) \\ &= \phi\left(\text{GroupConv}_{1\times 1}\left(h_c^f + h_{C-c+1}^b; G, \boldsymbol{\lambda}\right)\right) \end{aligned} \tag{24}$$

where $\phi(\cdot)$ represents the nonlinear activation function, G represents the number of groups, and $\boldsymbol{\lambda}$ represents the learnable parameters used to determine the group delimiters in Group Conv. The enhanced hidden states constitute a spatially aware bidirectional hidden state fusion sequence.

The fused state is then reinjected into subsequent forward and backward state evolution

$$\tilde{h}_c^f = h_c^{\text{fused}}, \qquad \tilde{h}_{C-c+1}^b = h_c^{\text{fused}}. \tag{25}$$

Finally, the spatially enhanced bidirectional states are mapped to the output sequences via the observation function

$$\begin{aligned} y_c^f &= \mathbf{C}^f \tilde{h}_c^f + \mathbf{D}^f x_c^f \\ y_c^b &= \mathbf{C}^b \tilde{h}_c^b + \mathbf{D}^b x_c^b \end{aligned} \tag{26}$$

yielding the bidirectional spectral feature representations

$$\begin{aligned} Y^f &= \left[y_1^f, y_2^f, \ldots, y_C^f\right] \\ Y^b &= \left[y_1^b, y_2^b, \ldots, y_C^b\right]. \end{aligned} \tag{27}$$

This design enables more sufficient modeling of spectral dependencies and spatial structural information within the state-space framework, thereby enhancing spectral–spatial joint representation for HSI classification.

V. Performance Evaluation

A. Datasets Description

We utilized four publicly available hyperspectral datasets (Indian Pines, Pavia University, WHU-Hi-LongKou, and WHU-Hi-HanChuan) to assess the discriminative ability of the proposed SAMamba model. All datasets employed a stratified random sampling strategy to partition annotated pixels into training and test sets. The numbers of training and test samples for each category are summarized in Table I.

1) Indian Pines Dataset: The Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) collected the data over the Indian Pines test site in northwestern Indiana, USA. The scene has a spatial dimension of 145 × 145 pixels with a spatial resolution of 20 m/pixel. It originally comprised 220 spectral bands covering 0.4–2.5 μm. After removing 20 water-absorption bands, 200 bands were retained for analysis. The ground truth data define 16 distinct land-cover classes, with 10% of labeled samples used for training.

2) Pavia University Dataset: The Reflective Optics System Imaging Spectrometer (ROSIS) captured this dataset during an aerial survey of Pavia, Italy. Comprising 610 × 340 pixels, the image exhibits a spatial resolution of 1.3 m/pixel. The original data contained 115 spectral bands spanning 0.43–0.86 μm, and 103 bands were retained after excluding 12 noisy or water-absorption bands. Nine land-cover categories are documented in the ground reference data. A 5% training ratio was adopted.

3) WHU-Hi-LongKou Dataset: This hyperspectral data were collected by a DJI Matrice 600 Pro (M600 Pro) uncrewed aerial vehicle (UAV) over Longkou Town, Hubei, China. The UAV platform carried a Headwall Nano-Hyperspec imaging sensor with an 8-mm focal length, yielding an HSI of 550×400 pixels with an approximate spatial resolution of 0.463 m. Covering 400–1000 nm across 270 bands, this dataset contains nine distinct land-cover types. Only 1% of labeled samples were used for training.

4) WHU-Hi-HanChuan Dataset: A Headwall Nano-Hyperspec imaging sensor was deployed on a Leica Aibot X6 UAV V1 platform to capture this dataset over Hanchuan, Hubei, China. Resulting in 1217 × 303 pixels, the acquired

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TABLE I
CATEGORY NAME AND SAMPLE NUMBERS OF TRAIN SET AND TEST SET OF EACH CLASS ON INDIAN PINES, PAVIA UNIVERSITY, WHU-HI-LONGKOU DATASETS, AND WHU-HI-HANCHUAN DATASETS

	Indian Pines			Pavia University			WHU-Hi-LongKou			WHU-HI-HanChuan		
No.	Category	Train	Test	Category	Train	Test	Category	Train	Test	Category	Train	Test
1	Alfalfa	5	41	Asphalt	332	6299	Corn	345	34166	Strawberry	447	44288
2	Corn-notill	143	1285	Meadows	932	17717	Cotton	84	8290	Cowpea	227	22526
3	Corn-mintill	83	747	Gravel	105	1994	Sesame	30	3001	Soybean	102	10185
4	Corn	24	213	Trees	153	2911	Broad-leaf soybean	632	62580	Sorghum	53	5300
5	Grass-pasture	48	435	Painted metal sheets	67	1278	Narrow-leaf soybean	42	4109	Water spinach	12	1188
6	Grass-trees	73	657	Bare soil	251	4778	Rice	119	11735	Watermelon	45	4488
7	Grass-pasture-mowed	3	25	Bitumen	67	1263	Water	671	66385	Greens	59	5844
8	Hay-windrowed	48	430	Self-blocking bricks	184	3498	Roads and houses	71	7053	Trees	179	17799
9	Oats	2	18	Shadows	47	900	Mixed weed	52	5177	Grass	94	9375
10	Soybean-notill	97	875							Red roof	105	10411
11	Soybean-mintill	245	2210							Gray roof	169	16742
12	Soybean-clean	59	534							Plastic	36	3643
13	Wheat	20	185							Bare soil	91	9025
14	Woods	126	1139							Road	185	18375
15	Buildings	39	347							Bright object	11	1125
16	Stone	9	84							Water	754	74647
	Total	1024	9225	Total	2138	40638	Total	2046	202496	Total	2369	235270

imagery achieves a spatial resolution of 0.109 m and comprises 274 spectral bands spanning 400–1000 nm. This dataset encompasses 16 distinct land-cover categories, with 1% of labeled data used for training. Notably, the afternoon acquisition under low solar elevation angles introduced considerable shadow coverage throughout the scene.

B. Experimental Settings

1) Evaluation Metrics: The assessment of the proposed SAMamba model employed three evaluation metrics: overall accuracy (OA), defined as the percentage of correctly classified pixels; average accuracy (AA), the average of class-specific accuracies; and the Kappa (κ) coefficient, a measure of classification agreement. Superior performance is indicated by higher values on these scales.

2) Experimental Details: All experiments, including the proposed SAMamba and the baseline methods, were conducted in a PyTorch-based environment on an NVIDIA GeForce RTX 4060 Ti GPU with 16-GB VRAM. Unless otherwise specified, all methods were trained and evaluated under identical settings. Specifically, all models were optimized using Adam for 100 epochs with a batch size of 64 and an initial learning rate of 0.001, without data augmentation. For all methods, the input features were uniformly reduced to 30 dimensions by principal component analysis (PCA), and a fixed spatial–spectral patch size of 15×15 was used. In the Mamba component, the repetition number was set to 1, and the state dimension (D_{state}) was fixed to 1.

To ensure fairness and reproducibility, all methods shared exactly the same training and test splits, and the reported results were averaged over ten independent runs. For SAMamba, the key structural parameters were fixed across all experiments: the group number G in the BFS module was set to $\{1, 5, 15\}$ according to the adopted 15×15 patch size; the number of semantic clusters N in APC was determined by the category number of each dataset; and fixed subband mask sizes $k = \{3, 5\}$ were used in FAS to model complementary low- and mid-frequency structural components. Except for the unified training and evaluation settings above, all baseline architectures and method-specific hyperparameters strictly followed their original publications.

3) Comparative Experiments: To demonstrate its efficacy, the proposed SAMamba model was evaluated against four representative HSI classification categories: CNN-based model (SSRN [14]); Transformer-based methods (SF [23], SSFTT [25], and CASST [24]); and recent Mamba-based approaches (MambaHSI [36], IGroupssMamba [38], HSI-MFormer [39], SpiralMamba [37], DulaMamba [40], and S^{2}Mamba [41]).

C. Parameter Analysis

1) Different Input Patch Sizes: To alleviate the memory footprint in HSI processing, the input data were divided into patches. Fig. 8 investigates the effect of this preprocessing by testing sizes from 9×9 to 17×17 (in steps of 2). For the Indian and Pavia datasets, accuracy generally increases with patch size and reaches its peak at 15×15. A similar trend is also observed for the LongKou and HanChuan datasets. This indicates that smaller patches limit classification performance because of insufficient contextual information, while larger patches provide richer spatial context for each pixel sample and thus improve accuracy. However, overly large patches may introduce redundant noise and lead to information aliasing. Therefore, based on the above analysis, a uniform patch size of 15×15 is adopted for all datasets in this experiment.

2) Different Embedding Dims: As illustrated in Fig. 9, variations in the embedded dimensionality exert a significant influence on the core evaluation metrics (OA, AA, and Kappa) of the classification model across the four target datasets. Specifically, as the embedded dimensionality increases, all classification metrics generally show an upward trend. On the Indian dataset, the AA metric in particular exhibits a more pronounced improvement than OA and Kappa. However, when the embedded dimensionality exceeds 64, the improvement in AA, OA, and κ gradually saturates, and a decline in

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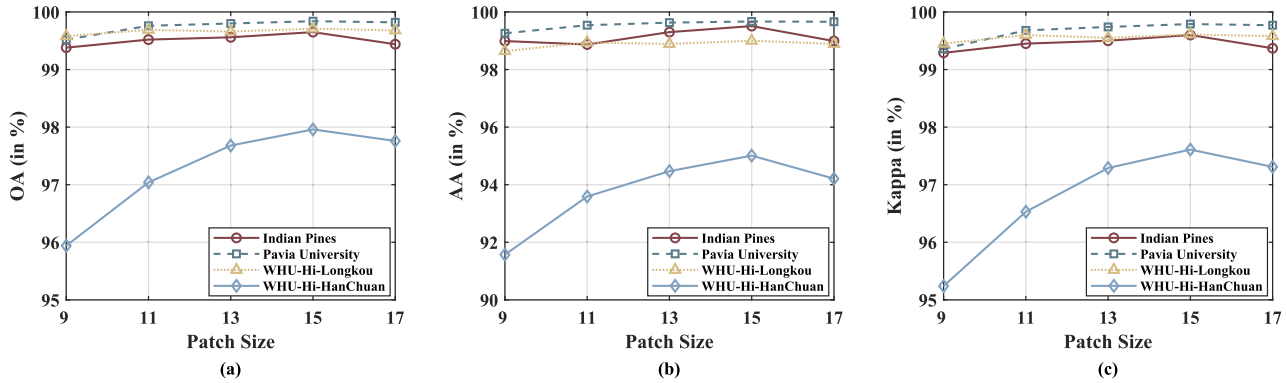


Fig. 8. Sensitivity analysis for the proposed method with a different feature patch size in terms of (a) OA, (b) AA, and (c) Kappa.

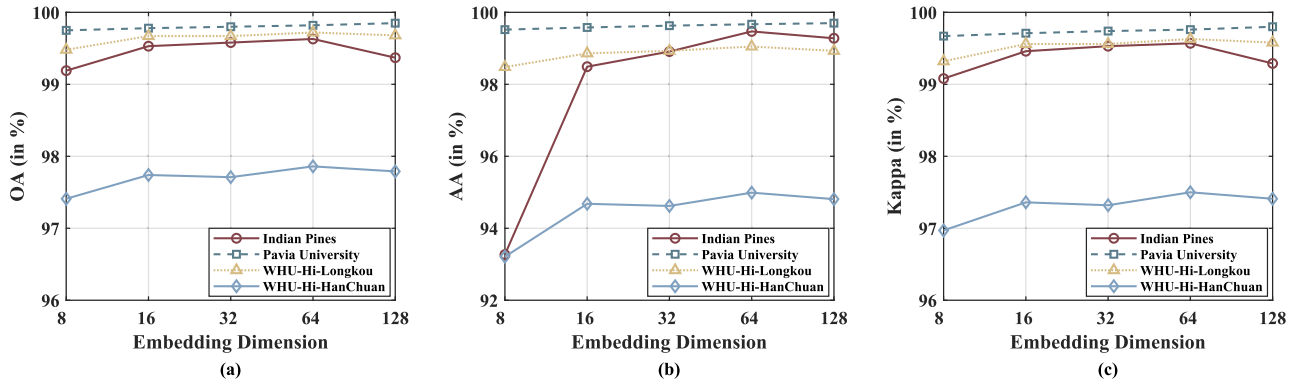


Fig. 9. Sensitivity analysis for the proposed method with different feature embedding dimensions in terms of (a) OA, (b) AA, and (c) Kappa.

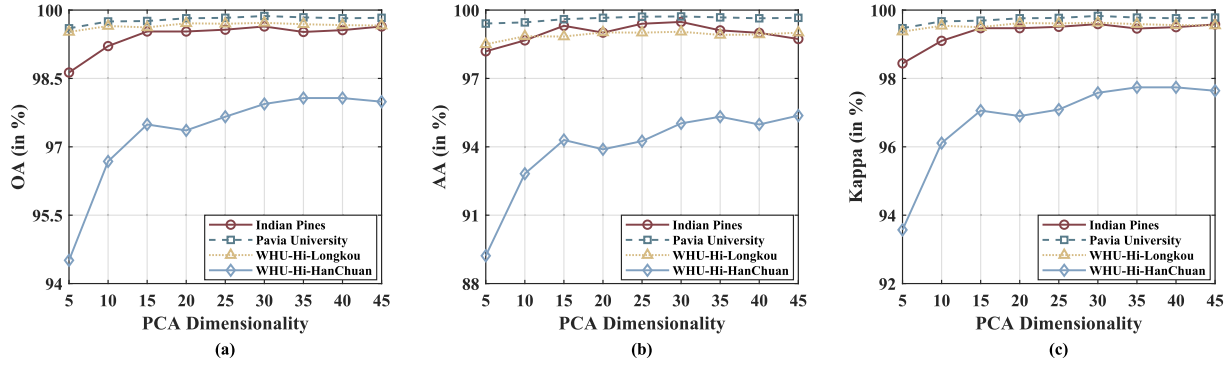


Fig. 10. Sensitivity analysis for the proposed method with a different feature PCA dimensionality in terms of (a) OA, (b) AA, and (c) Kappa.

accuracy is observed on most datasets, while the computational complexity continues to increase. Considering the tradeoff between classification performance and computational cost, an embedded dimensionality of 64 is finally identified as the optimal value.

3) Different Numbers of Principal Components in PCA: This study leverages PCA for feature reduction to alleviate the Hughes phenomenon caused by high-dimensional data and to reduce GPU memory usage, projecting the original data onto an orthogonal space that captures maximal variance. To determine the optimal number of components and examine its influence, configurations from $\{5, 10, \ldots, 45\}$ were evaluated. As shown in Fig. 10, all evaluation metrics first improve as the component number increases and achieve the best results at 30 components for most datasets. Beyond this value, the metrics begin to decline, indicating a pronounced Hughes effect. Therefore, considering both performance and efficiency, the first 30 principal components are selected as the optimal configuration.

D. Experimental Comparison With Competitive Approaches

1) Result on Indian Pines: Table II presents the classification results on Indian Pines dataset. The proposed method achieves the best performance in all three metrics, with OA,

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TABLE II

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR INDIAN PINES DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	CNN-based Method	Transformer-based Methods			Mamba-based Methods						Ours
	SSRN	SF	SSFTT	CASST	MambaHSI	IGroupSSMamba	HSI-MFormer	SpiralMamba	DualMamba	S^2Mamba	SAMamba
1	99.27 ± 1.56	95.12 ± 3.09	99.51 ± 1.46	96.59 ± 5.89	**100.00 ± 0.00**	98.29 ± 1.56	99.27 ± 1.12	99.27 ± 1.12	98.78 ± 1.22	**100.00 ± 0.00**	**100.00 ± 0.00**
2	99.11 ± 0.68	94.97 ± 1.63	98.93 ± 0.67	99.18 ± 0.54	**99.43 ± 0.18**	98.98 ± 0.67	99.13 ± 0.32	98.78 ± 0.38	99.35 ± 0.43	99.18 ± 0.54	99.36 ± 0.30
3	98.76 ± 1.22	93.12 ± 1.51	98.10 ± 1.32	97.78 ± 0.95	99.08 ± 0.45	98.31 ± 1.06	98.45 ± 0.80	98.23 ± 0.80	98.84 ± 1.33	99.21 ± 0.39	**99.64 ± 0.24**
4	98.59 ± 1.63	96.76 ± 1.69	99.20 ± 1.05	98.45 ± 1.44	99.25 ± 0.70	99.06 ± 0.63	99.77 ± 0.23	98.92 ± 0.87	99.67 ± 0.47	98.64 ± 0.90	**99.95 ± 0.14**
5	99.98 ± 0.07	97.78 ± 0.55	99.95 ± 0.14	99.79 ± 0.26	**100.00 ± 0.00**	99.66 ± 0.46	99.93 ± 0.11	99.91 ± 0.11	99.84 ± 0.29	99.98 ± 0.07	**100.00 ± 0.00**
6	98.71 ± 0.81	99.00 ± 0.43	98.42 ± 1.95	97.78 ± 1.82	98.74 ± 0.43	98.02 ± 0.66	98.95 ± 0.26	99.62 ± 0.31	98.69 ± 0.37	99.42 ± 0.49	**99.88 ± 0.11**
7	99.60 ± 1.20	83.20 ± 10.24	**100.00 ± 0.00**	99.20 ± 1.60	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.60 ± 1.20	99.60 ± 1.20	**100.00 ± 0.00**	**100.00 ± 0.00**
8	**100.00 ± 0.00**	99.95 ± 0.14	**100.00 ± 0.00**	99.74 ± 0.47	**100.00 ± 0.00**	99.79 ± 0.19	**100.00 ± 0.00**	**100.00 ± 0.00**	99.42 ± 0.61	**100.00 ± 0.00**	**100.00 ± 0.00**
9	84.44 ± 14.01	51.11 ± 6.48	94.44 ± 7.03	55.00 ± 23.89	**99.44 ± 1.67**	88.89 ± 12.42	96.67 ± 3.69	92.78 ± 5.58	82.78 ± 9.44	77.78 ± 6.57	96.67 ± 3.69
10	97.78 ± 1.18	92.54 ± 1.36	98.09 ± 1.10	96.30 ± 2.08	98.50 ± 0.32	95.83 ± 3.64	97.74 ± 0.54	97.55 ± 1.84	96.95 ± 0.49	98.26 ± 0.54	**99.55 ± 0.33**
11	98.62 ± 0.63	96.77 ± 0.64	98.41 ± 1.35	97.91 ± 0.69	98.92 ± 0.34	98.61 ± 0.65	98.70 ± 0.29	99.28 ± 0.41	99.07 ± 0.28	99.18 ± 0.34	**99.56 ± 0.15**
12	96.78 ± 1.03	87.87 ± 1.12	96.14 ± 1.54	95.11 ± 2.92	96.54 ± 1.35	94.76 ± 2.13	95.39 ± 0.53	96.76 ± 1.17	95.36 ± 2.04	97.27 ± 1.08	**98.24 ± 0.91**
13	99.57 ± 0.47	98.86 ± 0.51	99.14 ± 2.08	99.19 ± 1.73	97.03 ± 1.51	**99.73 ± 0.65**	99.62 ± 0.35	99.35 ± 0.53	98.86 ± 0.95	99.57 ± 0.68	**99.68 ± 0.55**
14	99.79 ± 0.33	99.16 ± 0.26	99.74 ± 0.27	99.22 ± 0.63	99.82 ± 0.15	99.64 ± 0.59	99.96 ± 0.06	99.68 ± 0.16	99.97 ± 0.06	99.96 ± 0.06	**99.99 ± 0.03**
15	99.22 ± 1.08	98.56 ± 0.94	98.99 ± 0.84	96.80 ± 2.66	99.31 ± 0.39	98.62 ± 0.94	98.85 ± 0.59	99.42 ± 0.52	99.51 ± 0.48	99.57 ± 0.30	**99.83 ± 0.19**
16	96.67 ± 5.24	98.45 ± 2.07	85.83 ± 9.74	64.64 ± 24.25	95.00 ± 1.67	93.93 ± 4.34	98.69 ± 1.45	97.74 ± 0.83	91.79 ± 14.26	98.69 ± 1.55	**99.40 ± 0.96**
OA (%)	98.80 ± 0.42	95.95 ± 0.32	98.55 ± 0.71	97.70 ± 0.71	98.98 ± 0.18	98.31 ± 0.38	98.80 ± 0.15	98.92 ± 0.25	98.74 ± 0.19	99.14 ± 0.18	**99.60 ± 0.07**
AA (%)	97.94 ± 1.28	92.70 ± 0.76	97.81 ± 1.11	93.30 ± 2.01	98.82 ± 0.22	97.63 ± 1.00	98.82 ± 0.35	98.56 ± 0.42	97.40 ± 1.16	97.92 ± 0.52	**99.48 ± 0.26**
$\kappa \times 100$	98.63 ± 0.48	95.38 ± 0.36	98.35 ± 0.81	97.38 ± 0.81	98.84 ± 0.21	98.08 ± 0.44	98.63 ± 0.17	98.77 ± 0.29	98.56 ± 0.21	99.02 ± 0.20	**99.54 ± 0.08**

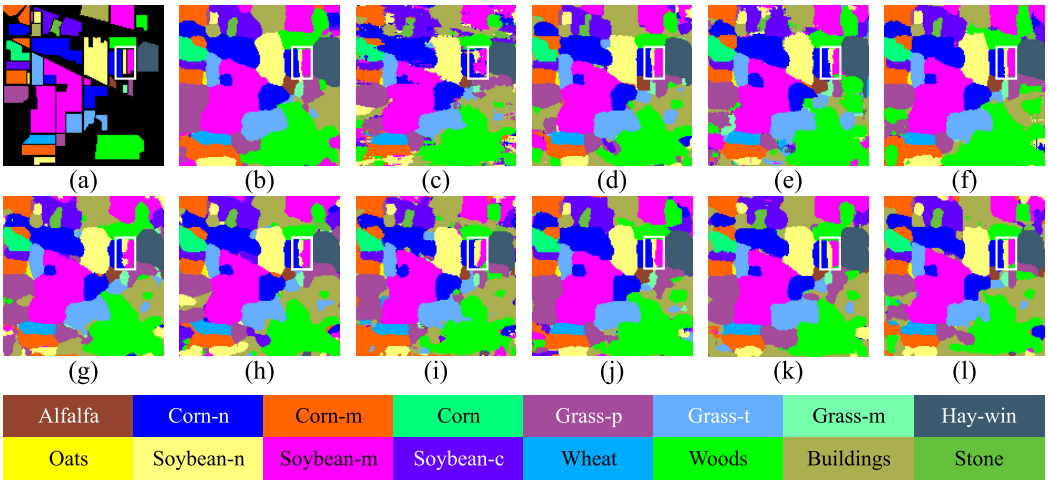


Fig. 11. Classification maps of different methods for Indian Pines dataset. (a) Ground truth. (b) SSRN. (c) SF. (d) SSFTT. (e) CASSST. (f) MambaHSI. (g) IGroupSSMamba. (h) HSI-MFormer. (i) SpiralMamba. (j) DualMamba. (k) S^2Mamba. (l) SAMamba.

AA, and κ reaching 99.60%, 99.48%, and 99.54%, respectively. Compared with the strongest baseline (S^2Mamba), it improves OA, AA, and κ by 0.46%, 1.56%, and 0.52%, respectively. At the class level, our method achieves the highest accuracy in 11 of 16 categories and 100% accuracy in four classes (Class 1: alfalfa, Class 5: grass-pasture, Class 7: grass-mowed, and Class 8: hay-windrowed), showing strong robustness in both homogeneous and complex regions.

Among the compared methods, the CNN-based SSRN achieves an OA of 98.80% and 100% accuracy in Class 8, but its local receptive field limits its performance compared with most Mamba-based models. For Transformer-based methods, SSFTT yields competitive results (OA: 98.55%), while SF performs worse due to insufficient spatial modeling. Among Mamba-based methods, S^2Mamba achieves the best baseline (OA: 99.14%) but is still outperformed by the proposed SAMamba, indicating more effective long-range spatial–spectral dependency modeling.

As shown in Fig. 11, SAMamba generates the cleanest classification map with fewer misclassifications and clearer boundaries. In contrast, other methods have obvious local errors and structural inconsistencies, especially in the marked region where SAMamba maintains better spatial consistency.

2) Results on the University of Pavia Dataset: Table III shows the classification results on the University of Pavia dataset. The proposed method achieves the best performance in all evaluation metrics, with an OA, AA, and Kappa coefficient of 99.85%, 99.71%, and 99.80%, respectively. Compared with the strongest baseline SpiralMamba, it improves OA, AA, and Kappa by 0.09%, 0.12%, and 0.11%, respectively. In terms of class-level accuracy, the proposed method reaches 100% accuracy in four of the nine land-cover classes, fully demonstrating its strong discriminative ability among different land-cover types.

Among comparison methods, the CNN-based SSRN achieves an OA of 99.51% and 100% accuracy for Class 6 (Bare soil). Among Transformer-based methods, SSFTT performs well with an OA of 99.57% and also achieves 100% accuracy for Class 6, while SF performs relatively poorly due to its overemphasis on spectral modeling. For Mamba-based approaches, SpiralMamba achieves the best baseline performance with an OA of 99.76%, and S^2Mamba also shows competitive performance with an OA of 99.74%. Despite their excellent classification results, these two Mamba-based methods are still outperformed by the proposed SAMamba.

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TABLE III

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE PAVIA UNIVERSITY DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	CNN-based Method	Transformer-based Methods			Mamba-based Methods						Ours
	SSRN	SF	SSFTT	CASST	MambaHSI	IGroupSSMamba	HSI-MFormer	SpiralMamba	DualMamba	S^2Mamba	SAMamba
1	99.87 ± 0.10	97.84 ± 0.38	99.89 ± 0.12	99.22 ± 1.79	99.87 ± 0.08	99.88 ± 0.09	99.48 ± 0.29	99.89 ± 0.11	99.96 ± 0.08	**99.96 ± 0.07**	**99.96 ± 0.07**
2	99.98 ± 0.02	99.98 ± 0.02	99.99 ± 0.01	99.98 ± 0.03	99.98 ± 0.02	**100.00 ± 0.00**	99.74 ± 0.79	99.99 ± 0.01	99.98 ± 0.03	99.98 ± 0.03	**100.00 ± 0.00**
3	99.29 ± 0.52	93.31 ± 1.22	99.21 ± 0.38	99.11 ± 0.66	98.69 ± 0.63	98.28 ± 2.60	97.69 ± 3.13	97.91 ± 1.28	98.96 ± 0.47	**99.63 ± 0.64**	99.36 ± 0.33
4	96.71 ± 0.90	98.07 ± 0.57	97.38 ± 0.37	97.36 ± 1.75	95.48 ± 0.38	97.57 ± 0.86	98.70 ± 0.37	**99.34 ± 0.28**	96.39 ± 0.72	97.57 ± 1.29	99.26 ± 0.31
5	99.78 ± 0.13	**99.99 ± 0.02**	99.56 ± 0.24	99.87 ± 0.14	99.86 ± 0.09	99.89 ± 0.12	**100.00 ± 0.00**	99.98 ± 0.07	99.61 ± 0.25	99.99 ± 0.02	**100.00 ± 0.00**
6	**100.00 ± 0.00**	99.65 ± 0.15	**100.00 ± 0.00**	99.95 ± 0.10	99.96 ± 0.02	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.87 ± 0.38	100.00 ± 0.01	**100.00 ± 0.00**
7	99.95 ± 0.10	99.69 ± 0.35	99.90 ± 0.16	99.49 ± 0.82	99.92 ± 0.04	99.99 ± 0.02	99.94 ± 0.07	**100.00 ± 0.00**	99.99 ± 0.02	**100.00 ± 0.00**	99.99 ± 0.02
8	**99.87 ± 0.20**	96.42 ± 0.58	99.40 ± 0.39	96.22 ± 5.18	98.20 ± 0.52	99.55 ± 0.29	99.25 ± 1.23	99.28 ± 0.45	99.87 ± 0.19	99.70 ± 0.36	99.43 ± 0.43
9	92.19 ± 2.30	97.84 ± 0.67	94.66 ± 1.63	98.60 ± 0.78	93.62 ± 0.91	93.64 ± 3.81	95.19 ± 2.31	99.91 ± 0.07	91.14 ± 8.34	98.87 ± 1.72	**99.39 ± 0.37**
OA (%)	99.51 ± 0.09	98.78 ± 0.08	99.57 ± 0.08	99.25 ± 0.57	99.27 ± 0.05	99.54 ± 0.18	99.43 ± 0.68	99.76 ± 0.09	99.44 ± 0.20	99.74 ± 0.13	**99.85 ± 0.04**
AA (%)	98.63 ± 0.27	98.09 ± 0.17	98.89 ± 0.21	98.87 ± 0.76	98.40 ± 0.10	98.76 ± 0.53	98.89 ± 0.84	99.59 ± 0.17	98.42 ± 0.90	99.52 ± 0.28	**99.71 ± 0.08**
$\kappa \times 100$	99.35 ± 0.12	98.39 ± 0.10	99.42 ± 0.10	99.01 ± 0.76	99.04 ± 0.06	99.39 ± 0.23	99.24 ± 0.89	99.69 ± 0.12	99.26 ± 0.27	99.66 ± 0.17	**99.80 ± 0.06**

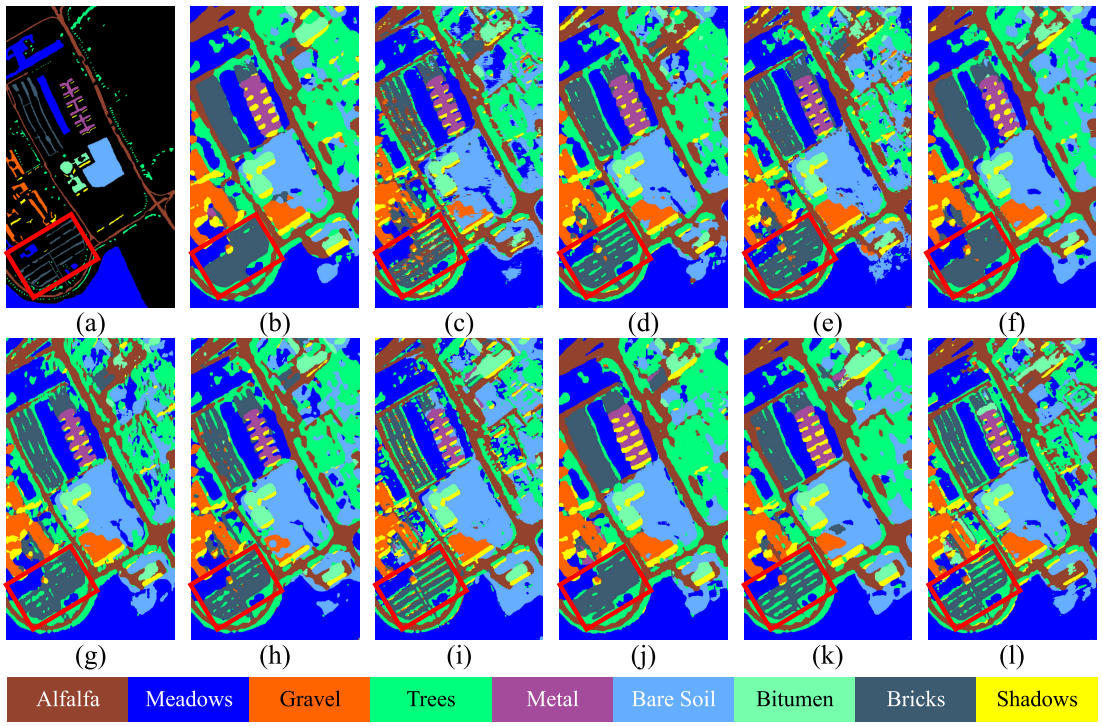


Fig. 12. Classification maps of different methods for the Pavia University dataset. (a) Ground truth. (b) SSRN. (c) SF. (d) SSFTT. (e) CASSST. (f) MambaHSI. (g) IGroupSSMamba. (h) HSI-MFormer. (i) SpiralMamba. (j) DualMamba. (k) S^2Mamba. (l) SAMamba.

TABLE IV

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE WHU-HI-LONGKOU DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	CNN-based Method	Transformer-based Methods			Mamba-based Methods						Ours
	SSRN	SF	SSFTT	CASST	MambaHSI	IGroupSSMamba	HSI-MFormer	SpiralMamba	DualMamba	S^2Mamba	SAMamba
1	99.92±0.11	99.99±0.01	99.82±0.45	99.92±0.07	99.96±0.01	99.98±0.02	99.99±0.01	**100.00±0.00**	99.80±0.46	99.96±0.02	99.99±0.01
2	99.15±1.19	97.28±0.97	99.51±0.41	99.00±0.64	99.17±0.35	97.70±4.48	98.76±0.31	99.43±0.17	99.19±0.33	98.74±0.08	**99.90±0.13**
3	98.86±1.76	96.90±1.28	98.51±2.94	98.39±1.50	99.03±0.74	98.38±0.49	98.70±0.33	99.50±0.25	99.34±0.36	99.69±0.03	**99.75±0.24**
4	99.52±0.72	99.49±0.11	99.66±0.23	99.33±0.38	99.59±0.09	99.60±0.39	99.79±0.06	99.69±0.07	99.59±0.11	99.49±0.02	**99.85±0.05**
5	96.29±4.56	90.89±1.29	96.99±2.04	95.55±1.85	97.59±0.55	97.93±0.60	97.10±0.40	97.71±0.73	97.55±0.57	97.14±0.20	**98.26±0.48**
6	99.13±0.84	99.69±0.11	99.61±0.23	96.93±1.03	99.43±0.14	98.98±0.60	99.69±0.05	**99.81±0.15**	98.63±0.30	99.10±0.23	99.74±0.19
7	99.84±0.33	99.96±0.01	99.92±0.05	99.89±0.12	99.92±0.02	99.97±0.02	99.97±0.00	**99.98±0.01**	99.92±0.03	99.96±0.01	**99.98±0.01**
8	94.90±2.23	96.56±0.86	96.68±0.91	92.35±1.27	95.80±0.41	96.73±1.45	97.33±0.33	**97.71±0.53**	94.32±1.13	96.01±1.46	97.63±0.26
9	92.39±2.22	93.46±1.53	92.84±1.98	86.38±4.82	95.23±0.77	94.50±3.69	95.12±0.54	**96.02±0.94**	94.49±1.26	93.31±1.90	95.93±0.90
OA (%)	99.24±0.41	99.18±0.08	99.41±0.14	98.79±0.27	99.44±0.05	99.39±0.38	99.56±0.02	99.62±0.04	99.30±0.10	99.35±0.03	**99.70±0.02**
AA (%)	97.78±0.92	97.14±0.33	98.17±0.55	96.41±0.58	98.41±0.17	98.20±0.84	98.50±0.03	98.87±0.17	98.09±0.23	98.15±0.02	**99.00±0.13**
$\kappa \times 100$	99.00±0.54	98.92±0.11	99.23±0.18	98.41±0.35	99.27±0.06	99.20±0.50	99.42±0.03	99.51±0.06	99.08±0.13	99.14±0.04	**99.60±0.03**

As shown in Fig. 12, the segmentation map generated by SAMamba is clearer, with fewer misclassified pixels and sharper boundaries. In contrast, other methods have obvious noise or blurred boundaries, especially in the red-boxed region, where SAMamba maintains better structural consistency and finer details.

3) Experimental Results on the WHU-Hi-LongKou Dataset: Table IV shows the classification results on the WHU-Hi-LongKou dataset. The proposed method achieves the optimal performance in all evaluation metrics, with OA, AA, and κ coefficient of 99.70%, 99.00%, and 99.60%, respectively. Compared with the strongest baseline (SpiralMamba), it

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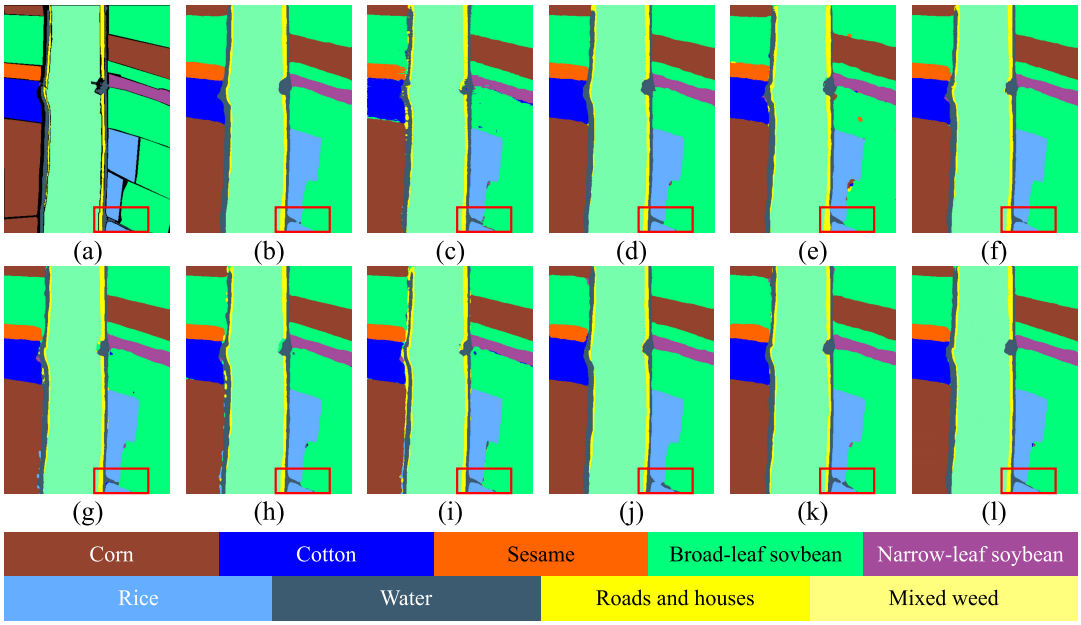


Fig. 13. Classification maps of different methods for the WHU-Hi-LongKou dataset. (a) Ground truth. (b) SSRN. (c) SF. (d) SSFTT. (e) CASSST. (f) MambaHSI. (g) IGroupSSMamba. (h) HSI-MFormer. (i) SpiralMamba. (j) DualMamba. (k) S^{2}Mamba. (l) SAMamba.

TABLE V

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE WHU-HI-HANCHUAN DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	CNN-based Method	Transformer-based Methods			Mamba-based Methods						Ours
	SSRN	SF	SSFTT	CASST	MambaHSI	IGroupSSMamba	HSI-MFormer	SpiralMamba	DualMamba	S^{2}Mamba	SAMamba
1	**99.34±0.65**	97.87±0.16	99.05±0.29	98.52±0.64	98.66±0.18	98.54±0.57	98.69±0.11	98.53±0.28	98.86±0.65	98.99±0.41	99.09±0.32
2	94.94±7.70	95.61±0.37	97.39±0.61	96.63±0.80	97.78±0.52	97.28±1.00	97.05±0.36	97.13±0.91	98.07±1.25	98.05±0.93	**97.96±0.87**
3	98.84±1.11	94.06±1.37	97.40±1.17	98.11±1.06	**99.06±0.44**	96.42±3.92	98.20±0.44	98.23±0.78	98.56±0.78	98.28±1.43	98.66±0.90
4	98.97±0.28	97.90±0.60	98.62±0.98	97.56±3.15	98.56±0.21	**99.11±0.21**	98.40±0.51	98.86±0.56	98.74±1.12	97.89±1.20	99.08±1.26
5	**94.71±3.06**	76.36±7.80	85.98±4.44	86.92±7.09	80.61±1.99	92.90±1.80	90.35±0.89	88.58±3.98	92.23±4.80	93.45±7.36	94.87±3.41
6	85.34±5.99	79.05±2.10	75.50±1.34	80.78±15.51	80.55±2.48	84.86±3.17	83.05±0.86	84.50±2.42	86.61±3.49	83.84±4.96	**87.61±3.36**
7	95.28±7.87	92.11±0.97	95.02±1.70	92.55±2.16	95.34±1.04	94.08±2.38	94.82±0.76	94.42±3.87	94.29±3.75	**96.12±1.33**	95.84±0.84
8	97.27±0.92	92.60±1.37	92.75±7.39	93.02±3.17	96.17±0.41	95.93±1.49	94.11±0.47	96.53±1.33	96.93±1.13	96.47±1.23	96.82±1.38
9	94.96±4.22	92.82±0.70	93.26±3.93	87.57±3.95	94.73±0.92	93.76±1.10	95.22±0.77	95.65±0.82	95.11±1.77	96.03±0.97	**96.36±1.37**
10	99.33±0.30	97.96±0.31	99.07±0.46	**99.17±0.49**	98.85±0.22	99.25±0.36	99.16±0.19	**99.35±0.20**	99.18±0.40	99.21±0.39	98.86±0.63
11	98.59±2.23	97.09±0.51	97.90±0.85	96.24±1.55	98.74±0.23	98.34±1.56	98.84±0.25	98.99±0.35	98.45±0.38	98.45±0.60	**99.06±0.26**
12	96.65±3.78	73.33±1.93	92.55±4.54	92.52±6.24	**98.98±0.87**	94.10±3.45	96.19±1.65	94.82±3.03	97.00±1.70	96.01±3.88	95.45±3.96
13	84.03±12.45	82.52±1.63	83.62±3.70	77.87±5.55	86.85±1.13	**89.44±2.42**	88.23±0.61	85.32±2.20	87.78±5.06	89.07±0.94	88.03±3.02
14	97.13±1.15	93.72±0.57	96.30±1.47	96.14±1.75	96.81±0.11	96.25±2.88	97.00±0.41	96.60±1.71	96.47±2.02	**97.67±0.48**	97.32±0.86
15	79.46±4.03	67.46±4.22	77.57±4.30	80.57±4.41	**83.33±3.76**	75.71±6.68	78.62±3.06	78.81±2.66	82.15±4.56	77.58±5.44	80.60±6.43
16	99.94±0.03	99.88±0.05	99.81±0.06	99.90±0.06	99.92±0.02	99.93±0.04	**99.96±0.02**	99.95±0.02	99.88±0.06	99.91±0.05	99.93±0.05
OA (%)	97.56±1.79	95.62±0.21	96.82±0.65	96.24±0.44	97.57±0.11	97.40±0.61	97.45±0.07	97.50±0.41	97.77±0.31	97.87±0.26	**98.00±0.31**
AA (%)	94.67±2.89	89.40±0.66	92.61±0.95	92.13±1.55	94.06±0.22	94.12±1.21	94.24±0.27	94.14±0.94	95.02±0.85	94.81±0.75	**95.35±0.83**
$\kappa \times 100$	97.14±2.10	94.87±0.24	96.28±0.77	95.60±0.51	97.15±0.13	96.96±0.71	97.02±0.08	97.07±0.48	97.39±0.37	97.51±0.31	**97.66±0.37**

improves OA, AA, and κ by 0.08%, 0.13%, and 0.09%, respectively. At the class level, the proposed method attains the highest accuracy in five out of nine categories and competitive results in the rest, demonstrating strong robustness for various land-cover types.

Among comparison methods, the CNN-based SSRN achieves an OA of 99.24%. For Transformer-based approaches, SSFTT performs well with an OA of 99.41%, followed by SF (99.18%). For Mamba-based methods, SpiralMamba achieves the best baseline performance (OA = 99.62%) and the highest accuracy in five categories (including 100% on Class 1), followed by HSI-MFormer (99.56%) and MambaHSI (99.94%). Despite its strong competitiveness, our model still outperforms SpiralMamba.

As shown in Fig. 13, the classification map of SAMamba is more precise with clearer boundaries and fewer misclassification regions. Specifically, in the red-boxed region, SpiralMamba and HSI-MFormer show inferior visual performance to our model, even with excellent quantitative metrics.

4) Experimental Results on the WHU-Hi-HanChuan Dataset: Table V shows the classification performance of various methods on the WHU-Hi-HanChuan dataset. The method proposed in this article achieves the best performance across all evaluation metrics, with an OA, AA, and κ coefficient of 98.00%, 95.35%, and 97.66%, respectively. Compared with the best-performing baseline method, S^{2}Mamba, the OA, AA, and κ coefficient improved by 0.13%, 0.54%, and 0.15%, respectively.

In terms of class-specific accuracy, the proposed method achieved the highest classification accuracy in 4 out of 16 classes, demonstrating its strong robustness in complex scenarios.

Comparing the baseline methods: the CNN-based SSRN method achieved an OA of 97.56%, demonstrating stable performance but falling short of most Mamba-based methods; among Transformer-based methods, SSFTT performed relatively well, with an OA of 96.82%. Among the Mamba-based methods, S^{2}Mamba achieved the best baseline performance

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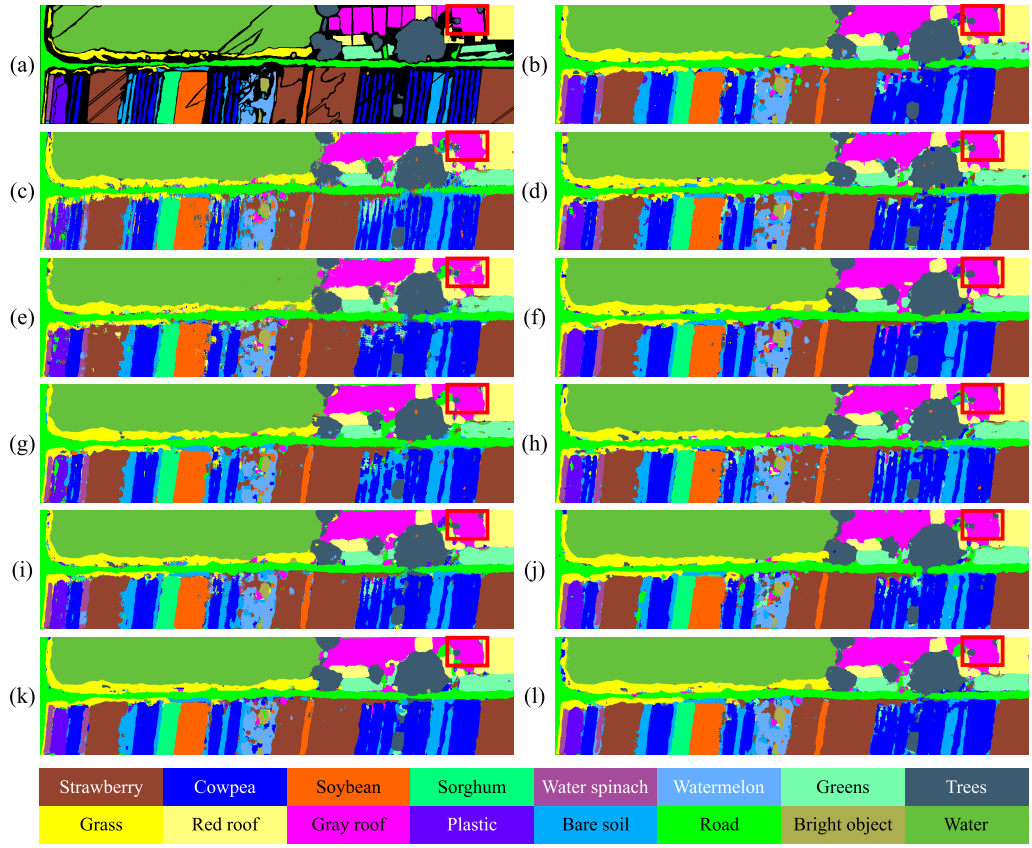


Fig. 14. Classification maps of different methods for the WHU-Hi-HanChuan dataset. (a) Ground truth. (b) SSRN. (c) SF. (d) SSFTT. (e) CASSST. (f) MambaHSI. (g) IGroupSSMamba. (h) HSI-MFormer. (i) SpiralMamba. (j) DualMamba. (k) S^2Mamba. (l) SAMamba.

(OA = 97.87%), followed by DualMamba (97.77%) and MambaHSI (97.57%). The proposed SAMamba further achieved a performance breakthrough, demonstrating its ability to more effectively model space-spectral dependencies in complex, shadow-covered environments.

Fig. 14 shows the classification visualization results for different models. It can be seen that SAMamba achieves higher classification accuracy, produces less prediction noise, and yields clearer object boundaries. The comparison methods exhibit significant misclassification in the shadowed area marked by the red box, whereas SAMamba maintains excellent classification performance in this region.

E. Statistical Significance Analysis

All statistical significance analyses are conducted based on ten independent runs with different random seeds, where the training–testing splits follow the same protocol as in the previous experimental settings. Paired t-tests are employed to compare SAMamba with the corresponding second-best competing method on each dataset under identical settings.

As reported in Table VI, SAMamba consistently achieves statistically significant improvements over the second-best methods across all four datasets in terms of OA, AA, and $\kappa \times 100$. On the Indian Pines dataset, the improvements are extremely significant ($p < 0.001$), demonstrating clear superiority even under the near-saturated accuracy setting.

For the Pavia University dataset, SAMamba also yields significant gains on all evaluation metrics. On the WHU-Hi-LongKou dataset, significant improvements are observed on OA and $\kappa \times 100$, while the improvement on AA is marginally significant. On the WHU-Hi-HanChuan dataset, SAMamba again outperforms the second-best method on all metrics with statistically significant advantages. These results confirm that the performance gains of SAMamba are significant and consistent across different datasets and evaluation criteria.

F. Ablation Study

To validate the proposed method, we conducted comprehensive ablation studies, with quantitative results reported in Table VII. These ablations are designed not only to verify the optimal combination but also to examine whether each module contributes to its intended role in sequence construction or state evolution. We further analyzed the model using t-SNE feature visualizations (see Fig. 15) and performance histograms.

At the architecture level, the full model (APC+FAS+BSS+BFS) consistently outperformed the variants with only the spatial branch (APC+FAS) or only the spectral branch (BSS+BFS) on all datasets. Its t-SNE results also showed the clearest category separation. These results demonstrate that the dual-path parallel architecture effectively captures complementary spatial and spectral information, which is essential for modeling complex spectral–spatial interactions.

For branch design, removing the FAS module from the spatial branch caused a clear performance drop, indicating

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TABLE VI

STATISTICAL SIGNIFICANCE ANALYSIS BETWEEN SAMAMBA AND THE SECOND-BEST COMPETING METHOD ON FOUR HYPERSPECTRAL DATASETS. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Dataset	Metric	SAMamba	Second-best result	Second-best method	p-value
Indian Pines	OA (%)	99.58 ± 0.10	99.10 ± 0.23	S^2Mamba	2.7×10^{-4}
	AA (%)	99.33 ± 0.38	98.04 ± 0.62	S^2Mamba	1.9×10^{-4}
	$\kappa \times 100$	99.52 ± 0.12	98.98 ± 0.26	S^2Mamba	2.7×10^{-4}
Pavia University	OA (%)	99.85 ± 0.07	99.77 ± 0.10	SpiralMamba	2.4×10^{-2}
	AA (%)	99.71 ± 0.12	99.59 ± 0.18	SpiralMamba	3.5×10^{-2}
	$\kappa \times 100$	99.81 ± 0.09	99.69 ± 0.13	SpiralMamba	2.4×10^{-2}
WHU-Hi-LongKou	OA (%)	99.68 ± 0.06	99.61 ± 0.06	SpiralMamba	7.6×10^{-3}
	AA (%)	99.05 ± 0.24	98.94 ± 0.16	SpiralMamba	1.0×10^{-1}
	$\kappa \times 100$	99.58 ± 0.07	99.49 ± 0.08	SpiralMamba	7.7×10^{-3}
WHU-Hi-HanChuan	OA (%)	98.06 ± 0.18	97.87 ± 0.29	S^2Mamba	3.2×10^{-2}
	AA (%)	95.27 ± 0.77	94.80 ± 0.80	S^2Mamba	1.1×10^{-1}
	$\kappa \times 100$	97.73 ± 0.21	97.50 ± 0.34	S^2Mamba	3.2×10^{-2}

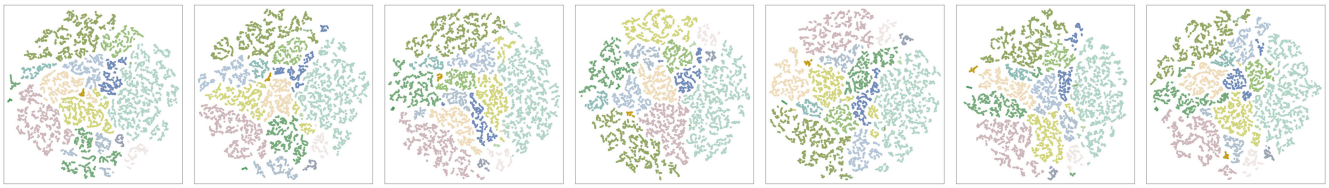


Fig. 15. t-SNE visualization of feature representations aggregated through different ablation strategies. (a) APC+FAS+BSS+BFS. (b) APC+FAS. (c) BSS+BFS. (d) APC+FAS+BFS. (e) FAS+BSS+BFS. (f) APC+FAS+BSS. (g) APC+BSS+BFS.

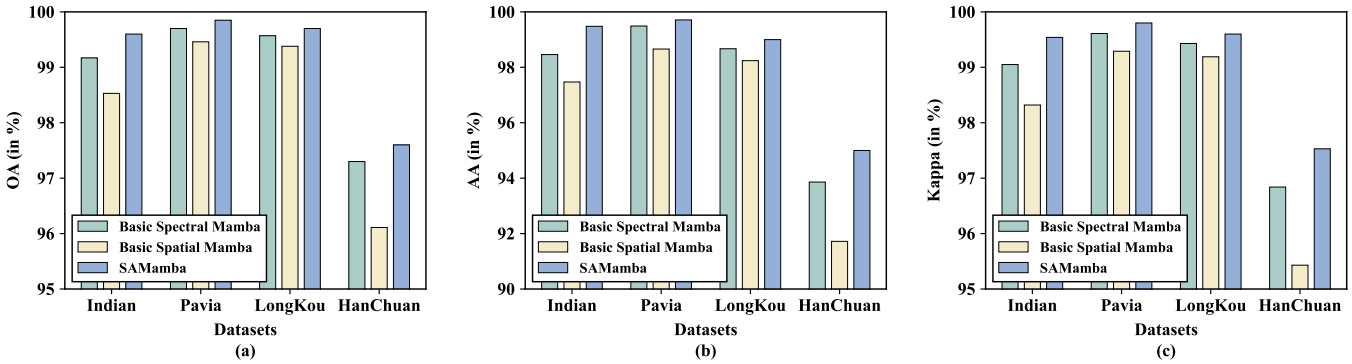


Fig. 16. Classification accuracy of basic spatial Mamba and spectral Mamba. (a) OA, (b) AA, and (c) Kappa across four datasets.

TABLE VII

ABLATION STUDY RESULTS (MEAN ± STANDARD DEVIATION)

Module Configuration				Indian Pines			Pavia University			WHU-Hi-LongKou			WHU-Hi-HanChuan		
APC	FAS	BSS	BFS	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
✓	✓			99.11 ± 0.24	97.60 ± 0.77	98.99 ± 0.28	99.81 ± 0.05	99.57 ± 0.16	99.74 ± 0.07	99.55 ± 0.07	98.60 ± 0.36	99.40 ± 0.09	97.77 ± 0.51	94.47 ± 1.03	97.39 ± 0.59
		✓	✓	99.51 ± 0.11	99.15 ± 0.37	99.44 ± 0.13	99.82 ± 0.06	99.66 ± 0.12	99.77 ± 0.08	99.69 ± 0.06	98.96 ± 0.23	99.59 ± 0.08	97.53 ± 0.60	94.31 ± 0.98	97.10 ± 0.70
✓	✓	✓		99.38 ± 0.18	98.83 ± 0.64	99.29 ± 0.21	99.80 ± 0.02	99.64 ± 0.03	99.73 ± 0.03	99.65 ± 0.16	98.94 ± 0.36	99.54 ± 0.21	97.03 ± 1.28	93.92 ± 1.30	96.52 ± 1.49
✓	✓		✓	99.13 ± 0.89	98.75 ± 0.86	99.01 ± 1.01	99.61 ± 0.48	99.31 ± 0.72	99.48 ± 0.64	99.66 ± 0.12	98.89 ± 0.34	99.55 ± 0.15	97.77 ± 0.34	94.70 ± 0.70	97.39 ± 0.40
✓		✓	✓	99.49 ± 0.13	98.94 ± 0.48	99.42 ± 0.15	99.73 ± 0.32	99.58 ± 0.31	99.64 ± 0.43	99.59 ± 0.31	98.71 ± 0.87	99.46 ± 0.41	97.74 ± 0.29	94.77 ± 0.55	97.35 ± 0.33
	✓	✓	✓	99.22 ± 0.66	97.92 ± 3.44	99.11 ± 0.76	99.76 ± 0.10	99.57 ± 0.20	99.69 ± 0.13	99.59 ± 0.23	98.62 ± 0.79	99.46 ± 0.30	97.66 ± 0.40	94.50 ± 0.83	97.26 ± 0.47
✓	✓	✓	✓	**99.60 ± 0.09**	**99.48 ± 0.22**	**99.54 ± 0.11**	**99.85 ± 0.04**	**99.71 ± 0.09**	**99.80 ± 0.05**	**99.70 ± 0.08**	**99.00 ± 0.19**	**99.60 ± 0.10**	**98.06 ± 0.18**	**95.27 ± 0.77**	**97.73 ± 0.21**

that FAS helps to preserve spatial structural information during sequence modeling and works synergistically with APC to enhance long-range semantic interactions. In the spectral branch, omitting BSS also degraded performance, confirming its importance for modeling interband spectral dependencies. In addition, BFS alleviates the spatial information decoupling introduced by channel-independent processing by injecting spatial context into the spectral branch.

At the module level, removing APC led to a noticeable decline in accuracy, validating its effectiveness in constructing high-quality input sequences. Likewise, the performance reduction without BFS further highlights its critical

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TABLE VIII

ABLATION STUDY ON MODEL COMPLEXITY AND EFFICIENCY (PARAMS/FLOPS ARE DATASET-INVARIANT)

Module Configuration				Complexity		Indian Pines		Pavia University		WHU-HI-LongKou		WHU-HI-HanChuan	
APC	FAS	BSS	BFS	Params (M)	FLOPs (G)	Training (s)	Inference (s)	Training (s)	Inference (s)	Training (s)	Inference (s)	Training (s)	Inference (s)
✓	✓			0.1850	0.0247	12.77	0.56	24.74	2.08	23.36	10.03	29.77	12.83
		✓	✓	0.1930	0.0205	12.73	0.45	24.64	1.68	23.22	8.13	29.78	10.41
✓	✓	✓		0.1850	0.0247	15.54	0.63	28.31	2.24	27.11	10.96	34.77	13.99
✓	✓		✓	0.2498	0.0288	14.63	0.57	30.18	2.36	28.80	11.40	37.00	14.72
✓		✓	✓	0.2482	0.0285	14.70	0.59	28.70	2.09	27.05	10.17	35.30	12.94
	✓	✓	✓	0.1945	0.0208	13.05	0.50	25.69	1.87	24.18	8.99	31.50	11.64
✓	✓	✓	✓	**0.2498**	**0.0288**	**16.64**	**0.66**	**32.51**	**2.47**	**30.71**	**12.14**	**39.64**	**15.34**

TABLE IX

COMPARISON OF DIFFERENT SPATIAL SEQUENCE CONSTRUCTION STRATEGIES (MEAN ± STANDARD DEVIATION)

Scan Strategy	Indian Pines			Pavia University			WHU-Hi-LongKou			WHU-Hi-HanChuan		
	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
Raster Scan	98.67 ± 0.23	97.52 ± 0.28	98.48 ± 0.26	99.20 ± 0.55	98.34 ± 0.65	98.94 ± 0.73	99.41 ± 0.06	98.28 ± 0.21	99.22 ± 0.09	97.00 ± 0.31	94.12 ± 0.48	96.66 ± 0.36
Snake Scan	98.32 ± 0.95	96.86 ± 1.27	98.08 ± 1.08	99.27 ± 0.53	98.37 ± 0.59	99.03 ± 0.70	99.41 ± 0.07	98.24 ± 0.21	99.22 ± 0.09	96.75 ± 1.41	94.01 ± 2.72	96.46 ± 1.65
Spiral Scan	98.61 ± 0.30	98.09 ± 0.50	98.42 ± 0.35	99.43 ± 0.09	98.46 ± 0.23	99.24 ± 0.11	99.24 ± 0.27	97.52 ± 1.42	99.00 ± 0.35	96.54 ± 1.06	93.02 ± 2.44	96.22 ± 1.24
Local Scan	98.15 ± 1.56	96.76 ± 1.92	97.89 ± 1.79	99.44 ± 0.06	98.64 ± 0.14	99.26 ± 0.08	99.37 ± 0.04	98.22 ± 0.12	99.17 ± 0.06	97.10 ± 0.31	94.03 ± 0.62	96.74 ± 0.36
Sweep Scan	98.49 ± 0.55	96.50 ± 1.37	98.28 ± 0.62	99.43 ± 0.08	98.56 ± 0.18	99.25 ± 0.10	99.36 ± 0.07	97.97 ± 0.24	99.16 ± 0.09	97.11 ± 0.21	94.40 ± 0.35	96.73 ± 0.24
APC	98.81 ± 0.32	97.33 ± 1.24	98.65 ± 0.36	99.76 ± 0.05	99.55 ± 0.12	99.68 ± 0.07	99.46 ± 0.11	98.28 ± 0.29	99.29 ± 0.14	97.80 ± 0.33	94.37 ± 1.00	97.43 ± 0.39
APC + FAS	99.11 ± 0.24	97.60 ± 0.77	98.99 ± 0.28	99.81 ± 0.05	99.57 ± 0.16	99.74 ± 0.07	99.55 ± 0.07	98.60 ± 0.36	99.40 ± 0.09	97.97 ± 0.15	95.14 ± 0.33	97.62 ± 0.18
SAMamba	**99.60 ± 0.09**	**99.48 ± 0.22**	**99.54 ± 0.11**	**99.85 ± 0.04**	**99.71 ± 0.09**	**99.80 ± 0.05**	**99.70 ± 0.08**	**99.00 ± 0.19**	**99.60 ± 0.10**	**98.06 ± 0.18**	**95.27 ± 0.77**	**97.73 ± 0.21**

role in promoting spatial information flow within the spectral branch.

Furthermore, as shown in Fig. 16, the complete model consistently surpassed the two original Mamba baselines, namely, basic spectral Mamba and basic spatial Mamba. This further confirms the effectiveness of each proposed component and the superiority of their joint design, rather than a simple accumulation of independent modules.

G. Complexity and Efficiency Analysis of Ablation Configurations

To assess the computational overhead introduced by different module combinations, we report the model size (Params), FLOPs, and the measured training/inference time on four datasets in Table VIII. Note that Params and FLOPs are dataset-invariant, while runtime depends on data resolution and sampling.

As shown in Table VIII, all ablation variants remain lightweight, with parameter counts within 0.185M–0.250M and FLOPs within 0.020G–0.029G. The full model (APC+FAS+BSS+BFS) has 0.2498M parameters and 0.0288G FLOPs, which are slightly higher than those of most submodels, yet it remains compact overall.

In terms of efficiency, introducing more modules generally leads to a moderate increase in runtime. The full model incurs the highest cost, with inference times of 0.66 s on Indian Pines, 2.47 s on Pavia University, 12.14 s on WHU-HI-LongKou, and 15.34 s on WHU-HI-HanChuan. Despite being slower than simpler variants, the overhead remains limited, demonstrating a favorable accuracy–efficiency tradeoff.

H. Analysis of Spatial Sequence Construction Strategies

Table IX compares several scanning strategies, including raster scan, snake scan, spiral scan, local scan, sweep scan, as well as the APC and APC+FAS methods proposed in this article. The conventional strategies are all fixed-rule sequential scanning methods, whereas APC adaptively generates the scanning order according to feature semantics, and APC+FAS further alleviates the spatial structural bias introduced by state-space rearrangement.

Across all four datasets, APC+FAS consistently outperforms fixed scanning strategies, while the complete SAMamba achieves the best overall performance. This indicates that fixed scanning, which depends on predefined adjacency relationships, is limited in capturing long-range semantic correlations in HSIs. In contrast, the proposed method supports state propagation along semantically relevant paths and preserves spatial consistency through frequency-domain structural constraints, leading to more accurate spatio-spectral joint modeling. These results demonstrate the effectiveness of the proposed scanning mechanism and suggest that SSM-based visual modeling should move from fixed-path scanning toward adaptive scanning driven by both semantics and structure.

I. Layout Sensitivity Analysis

To evaluate the robustness of SAMamba with respect to architectural layout, we perform a layout sensitivity analysis from two aspects: branch interaction strategy and module placement relative to the SSM block. The quantitative results are reported in Table X.

Here, inside the SSM block refers to operations applied to the recurrent hidden state h before readout, whereas *outside the SSM block* denotes postreadout feature-level processing.

1) Branch Interaction Strategy: We compare two serial layouts (Spe→Spa and Spa→Spe) with the proposed parallel dual-branch design. Although both serial variants achieve competitive results, the parallel layout consistently yields the best OA, AA, and Kappa across all datasets. This indicates that parallel spectral–spatial modeling within the SSM block enables more effective interaction than sequential processing.

2) Module Placement: We further test variants in which FAS or BFS is moved outside the SSM block and applied only after readout. Although these variants still perform well, they show consistent degradation compared with the default

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TABLE X

LAYOUT SENSITIVITY ANALYSIS OF DIFFERENT HYBRID ARCHITECTURES WITH RESPECT TO BRANCH INTERACTION STRATEGY AND MODULE PLACEMENT WITHIN OR OUTSIDE THE SSM BLOCK

Architecture Layout	Indian Pines			Pavia University			WHU-Hi-LongKou			WHU-Hi-HanChuan		
	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
Serial (Spe→Spa)	99.18 ± 0.18	98.99 ± 0.40	99.07 ± 0.21	99.78 ± 0.08	99.59 ± 0.14	99.71 ± 0.10	99.49 ± 0.53	98.85 ± 0.17	99.33 ± 0.69	97.53 ± 0.39	94.21 ± 0.89	97.10 ± 0.46
Serial (Spa→Spe)	99.37 ± 0.48	99.19 ± 0.41	99.29 ± 0.55	99.83 ± 0.04	99.63 ± 0.06	99.77 ± 0.06	99.67 ± 0.04	98.93 ± 0.13	99.56 ± 0.05	97.68 ± 0.54	94.57 ± 1.12	97.28 ± 0.63
FAS outside SSM block (post-readout)	99.50 ± 0.23	99.07 ± 0.59	99.43 ± 0.26	99.78 ± 0.11	99.64 ± 0.13	99.71 ± 0.14	99.65 ± 0.16	98.91 ± 0.44	99.54 ± 0.21	97.76 ± 0.13	94.78 ± 0.40	97.37 ± 0.16
BFS outside SSM block (post-readout)	99.35 ± 0.19	98.80 ± 0.58	99.26 ± 0.21	99.78 ± 0.07	99.51 ± 0.17	99.71 ± 0.09	99.54 ± 0.06	98.56 ± 0.22	99.39 ± 0.08	97.56 ± 0.35	94.30 ± 0.92	97.14 ± 0.41
Parallel FAS+BFS inside SSM block	**99.60 ± 0.09**	**99.48 ± 0.22**	**99.54 ± 0.11**	**99.85 ± 0.04**	**99.71 ± 0.09**	**99.80 ± 0.05**	**99.70 ± 0.08**	**99.00 ± 0.19**	**99.60 ± 0.10**	**98.00 ± 0.31**	**95.35 ± 0.83**	**97.66 ± 0.37**

TABLE XI

COMPUTATIONAL COMPLEXITY ANALYSIS OF VARIOUS COMPARISON APPROACHES IN TERMS OF PARAMETERS, FLOPS, TRAINING TIME, AND TESTING TIME

Metrics	CNN-based Method	Transformer-based Methods			Mamba-based Methods						Ours
	SSRN	SF	SSFTT	CASST	MambaHSI	IGroupSSMamba	HSI-MFormer	SpiralMamba	DualMamba	S^2Mamba	SAMamba
Params (M)	0.2420	0.1414	0.1533	0.1460	0.4039	0.1118	0.2970	0.2353	0.2529	0.1288	0.2498
Flops (G)	0.5822	0.00508	0.01751	0.03663	0.04321	0.01160	0.08083	0.0344	0.0108	0.0011	0.0288
Training (s)	107.09	13.16	5.30	20.95	116.88	41.62	52.11	44.89	24.77	32.38	17.39
Inference (s)	1.25	0.56	0.24	0.67	3.25	0.90	1.39	0.67	0.67	0.82	0.74

design. This demonstrates that directly modulating the hidden state inside the SSM block is more effective than postreadout feature fusion.

Overall, while different layouts remain competitive, the proposed parallel dual-branch architecture with FAS and BFS integrated inside the SSM block delivers the most stable and superior performance.

J. Analysis of Computational Complexity

On the Indian Pines dataset, the computational efficiency of different methods is compared in Table XI in terms of Params, FLOPs, training time, and inference speed.

CNN-based method generally requires higher computational cost to achieve better performance, while Transformer-based methods are relatively lightweight and show fast inference. Mamba-based methods present a wider tradeoff range depending on the architecture design, with some models being lightweight but others introducing higher complexity.

Compared with all baselines, SAMamba achieves a well-balanced efficiency profile, with moderate parameters (0.2498M) and FLOPs (0.0288G). It also reduces training time (17.39 s) compared with most Mamba-based methods, while maintaining fast inference (0.74 s), indicating a favorable tradeoff between computational cost and efficiency.

K. Robustness Analysis Under Challenging Conditions

1) Few-Shot Robustness Analysis: To evaluate the robustness of SAMamba under limited-label scenarios, we conduct few-shot experiments by controlling the number of labeled training samples per class. Considering the limited samples in some classes of Indian Pines, we set the number of training samples per class to 5, 10, and 15. For Pavia University, WHU-Hi-LongKou, and WHU-Hi-HanChuan, the number is set to 10, 20, 30, 40, and 50 per class. All baselines are evaluated under the same sampling protocol, and OA is used as the evaluation metric.

As shown in Fig. 17, SAMamba achieves competitive or superior performance across different few-shot settings. In particular, when only a few labeled samples are available, SAMamba still outperforms most competing methods on the four datasets. These results demonstrate that the proposed sequence construction and state-space intervention mechanisms help learn stable and discriminative spectral–spatial representations from limited supervision.

TABLE XII

EFFECTIVENESS OF APC ON TRANSFORMER-BASED CASST (MEAN ± STANDARD DEVIATION)

Datasets	CASST Baseline			CASST + APC		
	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
Indian Pines	97.76±0.43	94.91±2.49	97.45±0.49	98.20±0.36	95.22±0.82	97.94±0.41
Pavia University	99.30±0.28	98.91±0.37	99.07±0.37	99.17±0.29	98.12±0.69	98.90±0.38
WHU-Hi-LongKou	98.60±0.45	95.80±1.86	98.15±0.59	99.06±0.09	97.23±0.34	98.76±0.12
WHU-Hi-HanChuan	96.26±0.71	92.47±1.03	95.62±0.84	97.13±0.34	93.76±0.69	96.64±0.40

2) Noise Robustness Analysis: To evaluate robustness under nonideal observation conditions, we add zero-mean Gaussian noise to the raw hyperspectral data during testing. All methods are trained on clean samples, while the test samples are corrupted by noise with different intensities, simulating deployment on degraded hyperspectral observations.

The noise intensity is controlled by the target signal-to-noise ratio (SNR), where a lower SNR indicates stronger noise. Specifically, 40 and 35 dB represent weak noise, 30 dB represents moderate noise, 25 and 20 dB represent strong noise, and 15 dB represents severe noise degradation. We use OA drop, defined as $\Delta\text{OA} = \text{OA}_{clean} - \text{OA}_{noisy}$, to evaluate noise robustness, where a smaller value indicates better stability under noise interference.

As shown in Fig. 18, the performance of all methods decreases as the SNR decreases. Nevertheless, SAMamba generally shows a smaller OA drop than most competing methods, especially under strong and severe noise conditions. This indicates that SAMamba can learn more noise-tolerant spectral–spatial representations, thereby improving robustness in degraded scenarios.

L. Generalizability of the Proposed Modules Across Architectures

Our method is based on an SSM, in which different modules exhibit distinct transfer capabilities. Since APC operates during the sequence construction phase and is decoupled from specific modeling mechanisms, it is highly adaptable and

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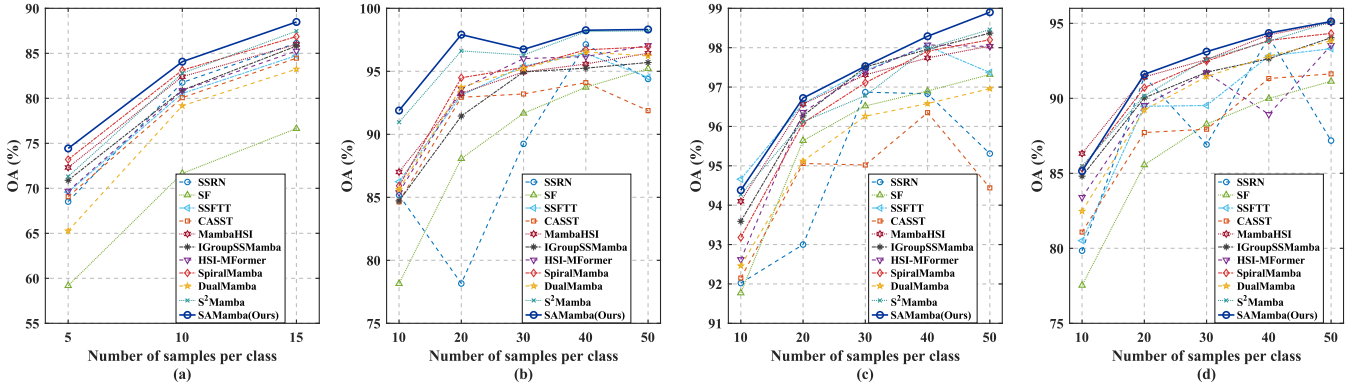


Fig. 17. OA variation under different few-shot training settings. (a) OA in Indian Pines. (b) OA in Pavia University. (c) OA in WHU-Hi-LongKou. (d) OA in WHU-Hi-HanChuan.

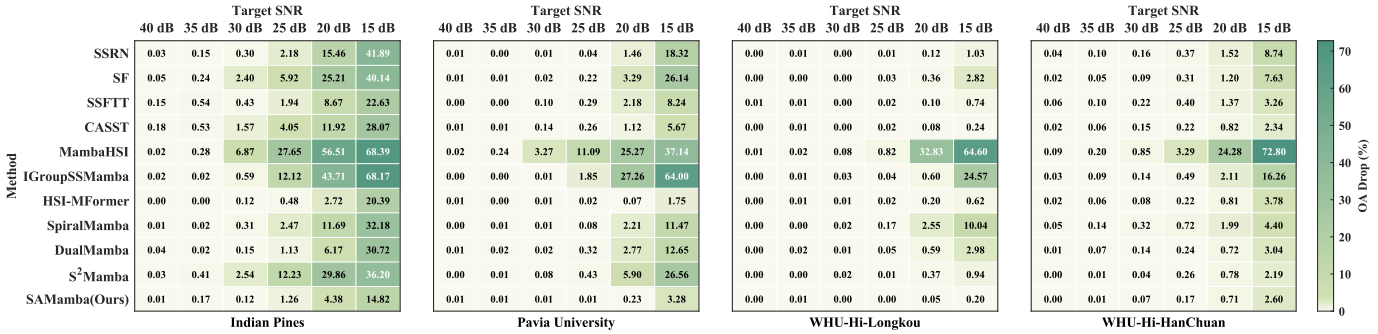


Fig. 18. OA drop under different SNR levels across four datasets.

can be directly transferred to the Transformer framework—its mechanism varies depending on the architecture: mitigating long-range dependency decay in Mamba and constructing semantic adjacencies to enhance attention interactions in Transformer. As shown in Table XII, integrating APC into CASST improved performance on three out of four datasets. A slight performance drop was observed on the University of Pavia dataset, suggesting that the benefits of APC may depend on the characteristics of the dataset and the baseline architecture. Overall, these results indicate that APC is not limited to the Mamba framework and demonstrates good transfer potential on attention-based architectures.

In contrast, FAS exhibits moderate scalability due to its frequency-domain-based structural modeling. However, BSS and BFS are tightly coupled with SSM's causal propagation and hidden-state representations, making direct transfer difficult. Overall, our method achieves strong cross-architecture generalization for key modules while preserving the advantages of SSM.

VI. Conclusion and Future Work

The SAMamba model is proposed to enhance Mamba with noncausal modeling ability similar to attention mechanisms. Specifically, APC constructs semantic sequences to improve interactions among semantically similar pixels, and FAS incorporates frequency-domain priors into the state-space updating process to strengthen spatial structural representation. Meanwhile, BSS builds bidirectional spectral sequences for comprehensive spectral context modeling, while BFS performs spatial-aware bidirectional hidden-state fusion to facilitate the joint evolution of spectral and spatial information. By integrating these modules, the proposed model mitigates the long-range dependency decay of causal sequence modeling and achieves unified spectral–spatial representation. Future work will investigate cross-dataset generalization and extend the model to super-resolution fusion applications for efficient large-scale image processing.

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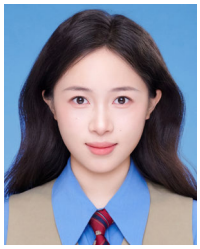
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